

stat_pn_weak_event_honest_inference

Specialization: gaussian_shell_directional_minimax_frontier

TL;DR. Tian and Wu’s identified PN/PS functionals, efficient influence functions, and cross-fitting architecture are taken as given. The contribution is an identical-shell achievability/converse pair for the nonlinear PN/PS ratio under known drifting factual-event schedules, with matched expected-diameter order $(nr_n)^{-1/2}$, plus a nonshrinking obstruction when $nr_n = O(1)$. The achievability proof now includes finite-sample Bernstein bounds for the explicit ridge-spline learners and a direct tensor-lattice inverse-row-sum theorem for their empirical Gram and logistic Hessian matrices.

1 Project justification

Gap. Regular PN/PS theory does not analyze triangular arrays with factual denominator $q_d(P_n) = r_n \rightarrow 0$. Generic weak-ratio and weak-instrument results provide nearby nonuniformity and inversion ideas, but not a matched expected-diameter frontier on one point-identified monotone PN/PS shell. The prior achievability sketch also left the spline inverse and concentration steps insufficiently justified; those steps are now discharged explicitly.

Niche. The note fixes a known factual-probability schedule and studies relative score and studentizer accuracy at variance scale $V_d(P_n) \asymp r_n$. PN uses smooth $e_n, m_{0,n}$, while PS uses smooth $e_n, m_{1,n}$. Local spline support, bounded design density, overlap, and the constrained propensity coefficient cube yield finite-sample learner control without importing opposite-direction outcome smoothness.

Fill. The explicit cross-fitted score inversion has uniform Gaussian coverage and expected diameter $O((nr_n)^{-1/2})$. The learner proof uses a written-out bounded-variable Bernstein inequality, an $O(J)$ -cardinality union bound, spectral stability of the local empirical Gram matrices, and a proved $O(J)$ inverse-row-sum bound. A causally valid two-point family in the identical shell supplies the matching lower order, while a separate bounded-count theorem gives nonshrinking diameter.

2 Related work

Tian and Wu provide the PN/PS identifying formulas, efficient influence functions, and cross-fitted regular inference. Cai and Kuroki provide regular variance-estimation context. Zhang–Wang–Miao study generally partially identified PN bounds, and Cuellar–Kennedy study regular inference for a projection of conditional probability of causation; neither target is the marginal monotonicity-identified PN/PS ratio considered here. Derumigny, Girard, and Guyonvarch analyze generic weak ratios, while Smucler, Lanni, and Masip and Ma develop identification-robust causal-ratio score inversion. Those works do not give the present identical-shell expected-diameter upper and lower pair at effective count nr_n . Low and Cai–Low supply the standard two-point confidence-length route; the new converse work is the causal same-shell construction. Cheng, Mao, Pearl, and Li study partially identified finite-discrete endpoints, a different estimand and regime. Hannart–Naveau climate attribution and Tian–Wu stroke attribution are conditional consumers after their factual schedules and shell conditions are justified.

3 Setup

Target (estimand / structural parameter): $\Theta_d(P_n)$, $d \in \{\text{PN}, \text{PS}\}$. Identifying functional / recovery map: $\Theta_{\text{PN}}(P) = 1 - E_P[e(X)m_0(X)]/E_P[e(X)m_1(X)]$, and $\Theta_{\text{PS}}(P) = 1 - E_P[(1 - e(X))(1 - m_1(X))]/E_P[(1 - e(X))(1 - m_0(X))]$.

- n — integer; $\{1, 2, \dots\}$; row index

Definition. n indexes a triangular-array row.

- p — integer; $\{1, 2, \dots\}$; dimension

Definition. p is the fixed covariate dimension.

- K — integer; $\{2, 3, \dots\}$; fold count

Definition. K is the fixed number of balanced cross-fitting folds.

- X — random vector; $[0, 1]^p$; covariate

Definition. X is the covariate vector.

- A — random variable; $\{0, 1\}$; treatment

Definition. A is the treatment indicator.

- Y — random variable; $\{0, 1\}$; outcome

Definition. Y is the observed outcome.

- O — random vector; $[0, 1]^p \times \{0, 1\}^2$; observed unit

Definition. $O = (X, A, Y)$.

- Z_n — random sample; $([0, 1]^p \times \{0, 1\}^2)^n$; observed row

Definition. $Z_n = (O_{1,n}, \dots, O_{n,n})$.

- P_n — probability law; observed-data laws; row law

Definition. P_n is the observed-data law in row n .

- P — probability law; observed-data laws; generic law

Definition. P is an individual observed-data law with its row index suppressed.

- $\tilde{P}_n$ — probability law; full-data laws; causal extension

Definition. $\tilde{P}_n$ is a law of $X, A, Y(0), Y(1)$ whose observed projection is P_n .

- $Y(0)$ — random variable; $\{0, 1\}$; potential outcome

Definition. $Y(0)$ is the control potential outcome.

- $Y(1)$ — random variable; $\{0, 1\}$; potential outcome

Definition. $Y(1)$ is the treated potential outcome.

- e_n — function; $[0, 1]^p \rightarrow [0, 1]$; propensity

Definition. $e_n(x) = P_n(A = 1 \mid X = x)$.

- $m_{0,n}$ — function; $[0, 1]^p \rightarrow [0, 1]$; control regression

Definition. $m_{0,n}(x) = E_{P_n}[Y \mid A = 0, X = x]$.

- $m_{1,n}$ — function; $[0, 1]^p \rightarrow [0, 1]$; treated regression

Definition. $m_{1,n}(x) = E_{P_n}[Y \mid A = 1, X = x]$.

- $f_{X,n}$ — function; Lebesgue densities on $[0, 1]^p$; design density

Definition. $f_{X,n}$ is a Lebesgue density of the P_n -marginal law of X on $[0, 1]^p$.

- ε — constant; $(0, 1/2]$; regularity constant

Definition. ε is the overlap bound.

- α — constant; $(0, 1/2)$; inference level

Definition. α is the nominal confidence-set error probability.

- $z_{1-\alpha/2}$ — constant; $\mathbb{R}$; critical value

Definition. $z_{1-\alpha/2}$ is the $1 - \alpha/2$ standard-normal quantile.

- B_ε — **constant**; $[1, \infty)$; **score envelope**

Definition. $B_\varepsilon = 2/\varepsilon - 1$ is the common fitted-score envelope.

- c_X — **constant**; $(0, \infty)$; **regularity constant**

Definition. c_X is the lower design-density bound.

- C_X — **constant**; $[c_X, \infty)$; **regularity constant**

Definition. C_X is the upper design-density bound.

- $\bar{d}$ — **constant**; $(0, 1/2]$; **LF witness cap**

Definition. $\bar{d}$ is the probability-simplex cap used only by least-favorable and bounded-count witness constructions.

- c_G — **constant**; $(0, \infty)$; **information constant**

Definition. c_G is the lower score-information constant.

- C_G — **constant**; $\{\varepsilon^{-1}\}$; **derived structural bound**

Definition. $C_G = \varepsilon^{-1}$ is the sharp derived monotone-overlap score-variance coefficient; it is not a free primitive.

- β_j — **constant**; $(0, \infty)$ for $j \in \{e, 0, 1\}$; **smoothness**

Definition. β_j is a Hölder smoothness index.

- L_j — **constant**; $(0, \infty)$ for $j \in \{e, 0, 1\}$; **smoothness**

Definition. L_j is a Hölder-ball radius.

- r_j — **integer**; $\{0, 1, \dots\}$ for $j \in \{e, 0, 1\}$; **sieve parameter**

Definition. r_j is a spline degree.

- s_j — **constant**; $(0, 1/2)$ for $j \in \{e, 0, 1\}$; **rate exponent**

Definition. $s_j = \beta_j / (2\beta_j + p)$.

- $a_{j,n}$ — **positive sequence**; $\mathbb{R}_+$ for $j \in \{e, 0, 1\}$; **spline rate**

Definition. For every integer $n \geq 1$, $a_{j,n} = \{\log(n \vee 3) / (n \vee 3)\}^{s_j} > 0$.

- d — **label**; $\{\text{PN}, \text{PS}\}$; **direction**

Definition. d indexes a causal direction.

- $b(d)$ — **map**; $\{\text{PN}, \text{PS}\} \rightarrow \{0, 1\}$; **regression index**

Definition. $b(\text{PN}) = 0$ and $b(\text{PS}) = 1$.

- r_n — **positive sequence**; $(0, 1]$; **Gaussian upper shell index**

Definition. r_n is a deterministic probability-valued directional factual-probability schedule; LF uses the separately capped subclass.

- $q_d(P)$ — **scalar functional**; $[0, 1]$; **factual probability**

Definition. $q_{\text{PN}}(P) = E_P[AY]$ and $q_{\text{PS}}(P) = E_P[(1 - A)(1 - Y)]$.

- $\Theta_d(P)$ — **scalar functional**; $[0, 1]$; **causal target**

Definition. $\Theta_d(P)$ is the direction- d functional in `def:functionals` when $q_d(P) > 0$.

- $t_d(P)$ — **scalar functional**; $[0, 1]$; **true target**

Definition. $t_d(P) = \Theta_d(P)$.

- $\eta^d(P)$ — nuisance tuple functional; functions on $[0, 1]^p$; score input

Definition. $\eta^{\text{PN}}(P) = (e, m_0)$ and $\eta^{\text{PS}}(P) = (1 - e, 1 - m_1)$.

- ψ_t^d — function; $([0, 1]^p \times \{0, 1\}^2) \times [0, 1] \rightarrow \mathbb{R}$; orthogonal score

Definition. ψ_t^d is the direction- d score in `def:scores` with the row index suppressed.

- $V_d(P)$ — scalar functional; $\mathbb{R}_+$; score variance

Definition. $V_d(P) = \text{Var}_P[\psi_{t_d(P)}^d(O; \eta^d(P))]$.

- $\mathsf{l}_d(P; c_G, C_G)$ — predicate; laws with $q_d(P) > 0$; information property

Definition. $\mathsf{l}_d(P; c_G, C_G)$ holds exactly when $V_d(P) \geq c_G q_d(P)$; its second displayed argument is the sharp derived coefficient $C_G = \varepsilon^{-1}$ and imposes no additional restriction.

- $\mathcal{B}_n$ — set; observed-data laws; LF witness and comparison class

Definition. $\mathcal{B}_n$ is the retained three-smoothness witness and comparison class in `def:base-class`; it is not a Gaussian upper parent.

- $\mathcal{R}^d$ — set; probability-valued positive sequences; Gaussian upper schedule class

Definition. $\mathcal{R}^d$ is the uncapped Gaussian-upper schedule class in `def:admissible-schedules`.

- $\mathcal{M}_{G,n}^d(r_n; c_G, C_G)$ — set; observed-data laws; Gaussian upper and identical-shell lower domain

Definition. $\mathcal{M}_{G,n}^d(r_n; c_G, C_G)$ is the generalized direction- d Gaussian shell defined by $q_d(P) = r_n$ and $V_d(P) \geq c_G r_n$ for any fixed $c_G > 0$; $C_G = \varepsilon^{-1}$ is only the sharp derived structural coefficient.

- $\mathcal{M}_{B,n}^d(r)$ — set; observed-data laws; bounded-count shell

Definition. $\mathcal{M}_{B,n}^d(r)$ is the direction- d bounded-count shell.

- $\mathcal{K}_{\text{LF}}$ — set; derived-compatible pairs; lower-information compatibility domain

Definition. $\mathcal{K}_{\text{LF}}$ contains the admissible lower-information constants paired with the sharp derived coefficient $C_G = \varepsilon^{-1}$.

- $\hat{g}_{d,n}$ — random function; $[0, 1] \rightarrow \mathbb{R}$; estimating equation

Definition. $\hat{g}_{d,n}$ is the cross-fitted direction- d score mean.

- $\hat{\sigma}_{d,n}^2$ — random function; $[0, 1] \rightarrow \mathbb{R}_+$; studentizer

Definition. $\hat{\sigma}_{d,n}^2$ is the held-out direction- d empirical score variance.

- $\mathcal{A}_{n,\alpha}^d$ — random set; subsets of $[0, 1]$; preliminary inversion

Definition. $\mathcal{A}_{n,\alpha}^d$ is the preliminary direction- d score-acceptance set.

- $\mathcal{C}_{n,\alpha}^{\text{score},d}$ — random set; subsets of $[0, 1]$; confidence set

Definition. $\mathcal{C}_{n,\alpha}^{\text{score},d}$ is the direction- d score-inversion set.

- D_n — random set; measurable subsets of $[0, 1]$; competitor set

Definition. D_n is an arbitrary measurable direction- d confidence set based on Z_n .

- $\mathcal{R}_{\text{LF}}^d$ — set; capped probability-valued positive sequences; LF Gaussian schedule class

Definition. $\mathcal{R}_{\text{LF}}^d$ is the least-favorable schedule subclass in `def:lf-admissible-schedules`.

- $\mathcal{B}_n^{\text{PN}}$ — set; observed-data laws; PN Gaussian upper parent

Definition. $\mathcal{B}_n^{\text{PN}}$ is the PN upper law class requiring smooth e_n and $m_{0,n}$, but not smooth $m_{1,n}$.

- $\mathcal{B}_n^{\text{PS}}$ — **set**; observed-data laws; PS Gaussian upper parent

Definition. $\mathcal{B}_n^{\text{PS}}$ is the PS upper law class requiring smooth e_n and $m_{1,n}$, but not smooth $m_{0,n}$.

- $\bar{\mathcal{C}}_{n,\alpha}^d$ — **random set**; nonempty compact subsets of $[0, 1]$; **propensity-only confidence set**

Definition. $\bar{\mathcal{C}}_{n,\alpha}^d$ is the uncapped propensity-only score-inversion set in `def:boundary-score-inversion`.

- B_e — **constant**; $(0, \infty)$; **propensity logit coefficient bound**

Definition. $B_e = 1 + \log\{(1 - \varepsilon)/\varepsilon\} + C_{\text{QI}}$, where $C_{\text{QI}} < \infty$ is the fixed coefficient-deviation constant of the chosen degree- r_e tensor-product B-spline quasi-interpolation rule on the Hölder class obtained by applying the logit map to $\mathcal{H}^{\beta_e}(L_e) \cap [\varepsilon, 1 - \varepsilon]$. The constant depends only on $\varepsilon, \beta_e, L_e, p, r_e$, never on n or the sample.

4 Assumptions

Assumption 1 (ass:iid-row). $Z_n \sim P_n^{\otimes n}$.

Standard (*row-wise independent sampling*, [5]).

Assumption 2 (ass:consistency). $\tilde{P}_n\{Y = Y(A)\} = 1$.

Standard (*consistency*, [1]).

Assumption 3 (ass:conditional-exchangeability). $(Y(0), Y(1)) \perp_{\tilde{P}_n} A \mid X$.

Standard (*conditional exchangeability*, [2]).

Assumption 4 (ass:monotonicity). $\tilde{P}_n\{Y(1) \geq Y(0)\} = 1$.

Standard (*outcome monotonicity*, [1]).

Assumption 5 (ass:overlap). $P_n\{\varepsilon \leq e_n(X) \leq 1 - \varepsilon\} = 1$ for fixed $\varepsilon \in (0, 1/2]$.

Standard (*uniform overlap*, [5]).

Assumption 6 (ass:design-density). For every row n , $c_X \leq f_{X,n}(x) \leq C_X$ for Lebesgue-almost every $x \in [0, 1]^p$, where the fixed constants satisfy $0 < c_X \leq C_X < \infty$ and do not depend on n .

Standard (*bounded design density*, [6]).

Assumption 7 (ass:holder-propensity). $e_n \in \mathcal{H}^{\beta_e}(L_e)$ for fixed $\beta_e, L_e > 0$.

Standard (*Hölder smoothness*, [6]).

Assumption 8 (ass:holder-control). $m_{0,n} \in \mathcal{H}^{\beta_0}(L_0)$ for fixed $\beta_0, L_0 > 0$.

Standard (*Hölder smoothness*, [6]).

Assumption 9 (ass:holder-treated). $m_{1,n} \in \mathcal{H}^{\beta_1}(L_1)$ for fixed $\beta_1, L_1 > 0$.

Standard (*Hölder smoothness*, [6]).

Assumption 10 (ass:rarity-cap). $0 < \bar{d} \leq 1/2$.

Novel. This construction-specific cap makes the capped monotone family a probability model: in `def` : least – favorable, the type mass $1 - 2r$ must be nonnegative, so $r \leq 1/2$ is the probability-simplex cap.

Assumption 11 (ass:score-information). $\text{I}_d(P; c_G, C_G) \iff V_d(P) \geq c_G q_d(P)$.

Novel. Rare factual events can retain a nonvanishing fraction of score information when the causal contrast is interior. Only the lower nondegeneracy condition is primitive; the monotone binary structural class yields the upper bound mechanically.

5 Definitions

Definition 1 (def:holder). Defined object. *Hölder balls*

Construction. For $\beta > 0$, put $q_\beta = \lceil \beta \rceil - 1$ and $\gamma_\beta = \beta - q_\beta \in (0, 1]$. For a function $f : [0, 1]^p \rightarrow \mathbb{R}$ with continuous partial derivatives through order q_β , define the full Hölder norm by

$$\|f\|_{\mathcal{H}^\beta} = \sum_{|\nu| \leq q_\beta} \|D^\nu f\|_\infty + \max_{|\nu|=q_\beta} \sup_{x \neq x'} \frac{|D^\nu f(x) - D^\nu f(x')|}{\|x - x'\|_2^{\gamma_\beta}}.$$

Here $\nu \in \mathbb{N}_0^p$ is a multi-index, and $D^0 f = f$. Define

$$\mathcal{H}^\beta(L) = \{f : [0, 1]^p \rightarrow \mathbb{R} : \|f\|_{\mathcal{H}^\beta} \leq L\}.$$

Under this full-norm convention, a constant map $f(x) \equiv c$ has $\|f\|_{\mathcal{H}^\beta} = |c|$. Hence a Hölder ball contains every constant map with value in $[0, 1]$ exactly when $L \geq 1$.

Member properties. *ass:holder-propensity; ass:holder-control; ass:holder-treated.*

Definition 2 (def:functionals). Defined object. *identified directional functionals*

Construction. $\Theta_{\text{PN}}(P) = 1 - E_P[e(X)m_0(X)]/E_P[e(X)m_1(X)]$ and $\Theta_{\text{PS}}(P) = 1 - E_P[(1 - e(X))(1 - m_1(X))]/E_P[(1 - e(X))(1 - m_0(X))]$, each defined only when its denominator is positive.

Inputs. $P; e; m_0; m_1$.

Definition 3 (def:scores). Defined object. *directional orthogonal scores*

Construction. $\psi_{t,n}^{\text{PN}}(O; \eta) = A\{Y - m_0(X)\} - e(X)\{1 - e(X)\}^{-1}(1 - A)\{Y - m_0(X)\} - tAY$. Define $\psi_{t,n}^{\text{PS}}(O; \eta^{\text{PS}}) = \psi_{t,n}^{\text{PN}}((X, 1 - A, 1 - Y); (1 - e, 1 - m_1))$. With the fitted inputs from $\text{def} : \text{learners}$, both scores satisfy $|\psi_t^d(O; \hat{\eta})| \leq B_\varepsilon = 2/\varepsilon - 1$ for $t \in [0, 1]$.

Inputs. $O; t; e; m_0; m_1; \varepsilon$.

Definition 4 (def:learners). Defined object. *fixed-fold ridge-regularized clipped spline learners*

Construction. Fix a deterministic balanced fold map $k(i) \in \{1, \dots, K\}$. For $j \in \{e, 0, 1\}$, set $s_j = \beta_j/(2\beta_j + p)$ and, for every integer $n \geq 1$, define

$$K_{j,n} = \begin{cases} 1, & n < 3, \\ \lceil (n/\log n)^{1/(2\beta_j+p)} \rceil, & n \geq 3, \end{cases} \quad J_{j,n} = (K_{j,n} + r_j + 1)^p, \quad \lambda_{j,n} = n^{-2}, \quad a_{j,n} = \left\{ \frac{\log(n \vee 3)}{n \vee 3} \right\}^{s_j} > 0,$$

with $r_j \geq \lceil \beta_j \rceil$. Let $B_{j,n}(x) \in \mathbb{R}^{J_{j,n}}$ be the nonnegative partition-of-unity tensor-product B-spline basis of degree r_j on the deterministic quasi-uniform grid with $K_{j,n}$ cells per coordinate. For the fixed constant B_e , define the compact convex coefficient cube

$$\Theta_{e,n} = \{\theta \in \mathbb{R}^{J_{e,n}} : \|\theta\|_\infty \leq B_e\}.$$

On the training complement $I_{-k,n}$, with $N_{-k,n} = |I_{-k,n}|$, use the deterministic fallback $\hat{\theta}_{e,-k,n} = \hat{\theta}_{0,-k,n} = \hat{\theta}_{1,-k,n} = 0$ when $N_{-k,n} = 0$, and skip the next two minimizations. When $N_{-k,n} > 0$, define $\hat{\theta}_{e,-k,n}$ as the unique minimizer over $\Theta_{e,n}$ of

$$\frac{1}{N_{-k,n}} \sum_{i \in I_{-k,n}} [\log\{1 + \exp(B_{e,n}(X_{i,n})^\top \theta)\} - A_{i,n} B_{e,n}(X_{i,n})^\top \theta] + \lambda_{e,n} \|\theta\|_2^2.$$

For $a \in \{0, 1\}$, define $\hat{\theta}_{a,-k,n}$ as the unique minimizer over $\mathbb{R}^{J_{a,n}}$ of

$$\frac{1}{N_{-k,n}} \sum_{i \in I_{-k,n}} \mathbf{1}\{A_{i,n} = a\} \{Y_{i,n} - B_{a,n}(X_{i,n})^\top \theta\}^2 + \lambda_{a,n} \|\theta\|_2^2.$$

Set

$$\widehat{e}_{-k,n}(x) = \Pi_{[\varepsilon/2, 1-\varepsilon/2]} \left(\frac{\exp\{B_{e,n}(x)^\top \widehat{\theta}_{e,-k,n}\}}{1 + \exp\{B_{e,n}(x)^\top \widehat{\theta}_{e,-k,n}\}} \right), \quad \widehat{m}_{a,-k,n}(x) = \Pi_{[0,1]} \{B_{a,n}(x)^\top \widehat{\theta}_{a,-k,n}\}.$$

The propensity objective is continuous and strictly convex on the compact convex set $\Theta_{e,n}$, and each outcome objective is coercive and strictly convex, so every displayed minimizer exists uniquely, including for separated logistic samples and samples containing no observation from one treatment arm. The propensity minimizer is measurable as the pointwise limit of deterministic projected gradient descent from zero with a deterministic step below the reciprocal of the common gradient-Lipschitz bound; projection onto $\Theta_{e,n}$ is coordinatewise clipping and hence continuous. The outcome minimizers are measurable by deterministic gradient descent. Cross-fitting evaluates at observation i the learner trained on $I_{-k(i),n}$. Casewise evaluation of the clipped score gives $|\psi_t^d(O; \widehat{\eta})| \leq B_\varepsilon = 2/\varepsilon - 1$ and $|\psi_t^d(O; \widehat{\eta}) - E\psi_t^d(O; \widehat{\eta})| \leq 2B_\varepsilon$ for both directions and every $t \in [0, 1]$.

Inputs. Z_n ; K ; β_e ; β_0 ; β_1 ; r_e ; r_0 ; r_1 ; ε ; B_e .

Definition 5 (def:base-class). Defined object. *three-smoothness witness and comparison class*

Construction.

$B_n = \{P_n : P_n \text{ is the observed projection of a full-data law satisfying consistency, conditional exchangeability, monotonicity, overlap, the support-wide condition } c_X \leq f_{X,n}(x) \leq C_X \text{ for Lebesgue-almost every } x \in [0, 1]^p, \text{ and all three declared Hölder conditions}\}$

. This intersection class is retained for capped witnesses, bounded-count results, and internal comparisons; it is not the parent of either directional Gaussian upper shell.

Member properties. *ass:consistency*; *ass:conditional-exchangeability*; *ass:monotonicity*; *ass:overlap*; *ass:design-density*; *ass:holder-propensity*; *ass:holder-control*; *ass:holder-treated*.

Definition 6 (def:admissible-schedules). Defined object. *uncapped Gaussian-upper directional schedules*

Construction. For $d \in \{\text{PN}, \text{PS}\}$,

$$\mathcal{R}^d = \left\{ (r_n)_{n \geq 1} : 0 < r_n \leq 1, \quad \frac{r_n}{na_{e,n}^2 a_{b(d),n}^2} \longrightarrow \infty \right\},$$

where $b(\text{PN}) = 0$ and $b(\text{PS}) = 1$.

Inputs. r_n ; $a_{e,n}$; $a_{0,n}$; $a_{1,n}$.

Definition 7 (def:lf-compatible-domain). Defined object. *least-favorable-compatible lower-information domain*

Construction. $\mathcal{K}_{\text{LF}} = \{(c_G, C_G) : 0 < c_G \leq 3/16, C_G = \varepsilon^{-1}\}$, where the second coordinate records the sharp derived structural score-variance coefficient and is not a primitive information restriction. A primitive tuple is LF-compatible when $c_X \leq 1 \leq C_X$, overlap admits $e \equiv 1/2$, the declared Hölder balls contain $e \equiv 1/2$ and all constant regressions generated by *def:least-favorable* for $0 < r \leq \bar{d}$, and $0 < \bar{d} \leq 1/2$.

Inputs. c_X ; C_X ; ε ; c_G ; $\bar{d}$.

Definition 8 (def:gaussian-shells). Defined object. *direction-specific generalized Gaussian upper shells*

Construction. For fixed $c_G > 0$, set $C_G = \varepsilon^{-1}$. For $(r_n) \in \mathcal{R}^{\text{PN}}$,

$$\mathcal{M}_{G,n}^{\text{PN}}(r_n; c_G, C_G) = \{P \in \mathcal{B}_n^{\text{PN}} : q_{\text{PN}}(P) = r_n, V_{\text{PN}}(P) \geq c_G r_n\}.$$

For $(r_n) \in \mathcal{R}^{\text{PS}}$,

$$\mathcal{M}_{G,n}^{\text{PS}}(r_n; c_G, C_G) = \{P \in \mathcal{B}_n^{\text{PS}} : q_{\text{PS}}(P) = r_n, V_{\text{PS}}(P) \geq c_G r_n\}.$$

The coordinate C_G records the sharp derived structural upper coefficient and imposes no primitive restriction. LF compatibility and $\bar{d}$ are absent from both upper shells.

Member properties. *ass:score-information*.

Definition 9 (def:bounded-count-shell). Defined object. *directional bounded-count shell*

Construction. $\mathcal{M}_{B,n}^d(r) = \{P \in \mathcal{B}_n : q_d(P) = r\}$ for $d \in \{\text{PN}, \text{PS}\}$ and $0 < r \leq \bar{d}$.

Member properties. *ass:rarity-cap*.

Definition 10 (def:score-inversion). Defined object. *directional score-inversion set*

Construction. With fold map $k(i)$, set $\hat{g}_{d,n}(t) = n^{-1} \sum_{i=1}^n \psi_{t,n}^d(O_{i,n}; \hat{\eta}_{d,-k(i),n})$ and

$$\hat{\sigma}_{d,n}^2(t) = n^{-1} \sum_{i=1}^n [\psi_{t,n}^d(O_{i,n}; \hat{\eta}_{d,-k(i),n}) - \hat{g}_{d,n}(t)]^2$$

. Define

$$\mathcal{A}_{n,\alpha}^d = \{t \in [0, 1] : n\hat{g}_{d,n}(t)^2 \leq z_{1-\alpha/2}^2 \hat{\sigma}_{d,n}^2(t)\}$$

, and set $\mathcal{C}_{n,\alpha}^{\text{score},d} = \mathcal{A}_{n,\alpha}^d$ when $\mathcal{A}_{n,\alpha}^d \neq \emptyset$, and $\mathcal{C}_{n,\alpha}^{\text{score},d} = \{0\}$ otherwise. The score is affine in t and its empirical variance is continuous, so $\mathcal{A}_{n,\alpha}^d$ is a random closed subset of compact $[0, 1]$; the fallback gives a measurable nonempty compact-valued output and a defined measurable diameter. The fallback preserves coverage because

$$\{\Theta_d(P_n) \in \mathcal{A}_{n,\alpha}^d\} \subseteq \{\Theta_d(P_n) \in \mathcal{C}_{n,\alpha}^{\text{score},d}\}$$

, and cannot enlarge diameter. The common envelope B_ε bounds the held-out score and twice bounds its centered version, supporting both variance and score-set diameter control.

Inputs. $Z_n; \psi_{t,n}^d; \hat{\eta}_{d,-k,n}; \alpha; \varepsilon$.

Definition 11 (def:least-favorable). Defined object. *capped monotone least-favorable construction*

Construction. Let $X \sim \text{Unif}([0, 1]^p)$, let $A \sim \text{Bernoulli}(1/2)$ independently of $X, Y(0), Y(1)$, and set

$$P\{(Y(0), Y(1)) = (1, 1) \mid X\} = 2(1-t)r, \quad P\{(Y(0), Y(1)) = (0, 1) \mid X\} = 2tr, \quad P\{(Y(0), Y(1)) = (0, 0) \mid X\} = 1-2r$$

for $t \in [1/4, 3/4]$ and $0 < r \leq \bar{d}$, with observed outcome $Y = Y(A)$. This is the PN construction and has $e = 1/2$, $m_0 = 2(1-t)r$, and $m_1 = 2r$. Define the PS construction explicitly by

$$A^s = 1 - A, \quad Y^s(0) = 1 - Y(1), \quad Y^s(1) = 1 - Y(0), \quad Y^s = Y^s(A^s) = 1 - Y.$$

Its observed regressions are $e^s = 1/2$, $m_0^s = 1 - 2r$, and $m_1^s = 1 - 2(1-t)r$.

Inputs. $t; r; \bar{d}$.

Definition 12 (def:sharpness-handle). Defined object. *rare-learning sharpness handle*

Construction. Use a direction-specific regression perturbation with $L_2(P_X)$ size comparable to $a_{b(d),n}$ inside the monotone shell, then compare its score drift with $\sqrt{r_n/n}$ through a testing-affinity calculation.

Inputs. $r_n; a_{b(d),n}; \psi_{t,n}^d$.

Definition 13 (def:boundary-base-class). Defined object. *propensity-only monotone boundary class*

Construction. $\mathcal{B}_n^\partial$ is the set of observed-data laws P_n that are observed projections of full-data laws satisfying consistency, conditional exchangeability, monotonicity, overlap, and the support-wide bounded design-density condition, and whose propensity satisfies $e_n \in \mathcal{H}^{\beta_e}(L_e)$. The outcome regressions $m_{0,n}$ and $m_{1,n}$ are otherwise arbitrary measurable $[0, 1]$ -valued functions satisfying $m_{0,n}(X) \leq m_{1,n}(X)$ P_n -almost surely; no outcome-regression Hölder condition is imposed.

Member properties. *ass:consistency; ass:conditional-exchangeability; ass:monotonicity; ass:overlap; ass:design-density; ass:holder-propensity*.

Definition 14 (def:boundary-raw-shell). Defined object. *uncapped propensity-only raw boundary shell*

Construction. For $d \in \{\text{PN}, \text{PS}\}$, fixed $c_G > 0$, $0 < r \leq 1$, and row n ,

$$\mathcal{S}_{G,n}^{\partial,d}(r; c_G) = \{P \in \mathcal{B}_n^\partial : q_d(P) = r, V_d(P) \geq c_G r\}.$$

Member properties. *ass:score-information*.

Definition 15 (def:pn-upper-base-class). Defined object. *PN directional upper observed-law class*
Construction.

$\mathcal{B}_n^{\text{PN}} = \{P_n : P_n \text{ is the observed projection of a full-data law satisfying consistency, conditional exchangeability, monotonicity, overlap, } c_X \leq f_{X,n}(x) \leq C_X \text{ for Lebesgue-almost every } x \in [0, 1]^p, e_n \in \mathcal{H}^{\beta_e}(L_e), \text{ and } m_{0,n} \in \mathcal{H}^{\beta_0}(L_0)\}$
. No condition on the Hölder regularity of $m_{1,n}$ is imposed.

Member properties. *ass:consistency; ass:conditional-exchangeability; ass:monotonicity; ass:overlap; ass:design-density; ass:holder-propensity; ass:holder-control.*

Definition 16 (def:ps-upper-base-class). Defined object. *PS directional upper observed-law class*
Construction.

$\mathcal{B}_n^{\text{PS}} = \{P_n : P_n \text{ is the observed projection of a full-data law satisfying consistency, conditional exchangeability, monotonicity, overlap, } c_X \leq f_{X,n}(x) \leq C_X \text{ for Lebesgue-almost every } x \in [0, 1]^p, e_n \in \mathcal{H}^{\beta_e}(L_e), \text{ and } m_{1,n} \in \mathcal{H}^{\beta_1}(L_1)\}$
. No condition on the Hölder regularity of $m_{0,n}$ is imposed.

Member properties. *ass:consistency; ass:conditional-exchangeability; ass:monotonicity; ass:overlap; ass:design-density; ass:holder-propensity; ass:holder-treated.*

Definition 17 (def:lf-admissible-schedules). Defined object. *least-favorable capped Gaussian schedules*
Construction. For $d \in \{\text{PN}, \text{PS}\}$,

$$\mathcal{R}_{\text{LF}}^d = \{(r_n) \in \mathcal{R}^d : 0 < r_n \leq \bar{d} \text{ for every } n\}.$$

Inputs. $\mathcal{R}^d; \bar{d}$.

Definition 18 (def:propensity-learner). Defined object. *cross-fitted clipped ridge-spline propensity learner*

Construction. For each fold k , $\hat{e}_{-k,n}$ is the propensity component of **def:learners**: it is the clipped logistic tensor-product B-spline ridge minimizer trained on $I_{-k,n}$, with $K_{e,n}$, $J_{e,n}$, and $\lambda_{e,n}$ exactly as displayed there. This component is a measurable function only of $(X_i, A_i)_{i \in I_{-k,n}}$ and is unchanged if both outcome-regression minimizations are omitted.

Inputs. $Z_n; K; \beta_e; r_e; \varepsilon$.

Definition 19 (def:pn-learners). Defined object. *PN direction-specific cross-fitted learner*

Construction. For each fold k , define $\hat{\eta}_{\text{PN},-k,n} = (\hat{e}_{-k,n}, \hat{m}_{0,-k,n})$, where the first component is **def:propensity-learner** and the second is the clipped control-arm ridge least-squares tensor-product B-spline component displayed in **def:learners**. The two objectives are separable, so this pair is unchanged if the treated-regression objective is omitted. It uses no fitted m_1 .

Inputs. $Z_n; K; \beta_e; \beta_0; r_e; r_0; \varepsilon$.

Definition 20 (def:ps-learners). Defined object. *PS direction-specific cross-fitted learner*

Construction. For each fold k , define $\hat{\eta}_{\text{PS},-k,n} = (1 - \hat{e}_{-k,n}, 1 - \hat{m}_{1,-k,n})$, where $\hat{e}_{-k,n}$ is **def:propensity-learner** and $\hat{m}_{1,-k,n}$ is the clipped treated-arm ridge least-squares tensor-product B-spline component displayed in **def:learners**. The objectives are separable, so this tuple is unchanged if the control-regression objective is omitted. It uses no fitted m_0 .

Inputs. $Z_n; K; \beta_e; \beta_1; r_e; r_1; \varepsilon$.

Definition 21 (def:boundary-score-inversion). Defined object. *uncapped propensity-only affine score inversion*

Construction. Define

d	$U_d^\star(O)$	$J_d^\star(O)$
PN	AY	$(1 - A)Y$
PS	$(1 - A)(1 - Y)$	$A(1 - Y)$

and put $\hat{\pi}_{\text{PN},-k} = \hat{e}_{-k}$, $\hat{\pi}_{\text{PS},-k} = 1 - \hat{e}_{-k}$. Set

$$\bar{\phi}_{t,-k}^d(O) = (1-t)U_d^*(O) - \frac{\hat{\pi}_{d,-k}(X)}{1 - \hat{\pi}_{d,-k}(X)} J_d^*(O),$$

$$\bar{g}_{d,n}(t) = n^{-1} \sum_{i=1}^n \bar{\phi}_{t,-k(i)}^d(O_{i,n}), \quad \bar{\sigma}_{d,n}^2(t) = n^{-1} \sum_{i=1}^n \{\bar{\phi}_{t,-k(i)}^d(O_{i,n}) - \bar{g}_{d,n}(t)\}^2.$$

Let

$$\bar{\mathcal{A}}_{n,\alpha}^d = \{t \in [0, 1] : n\bar{g}_{d,n}(t)^2 \leq z_{1-\alpha/2}^2 \bar{\sigma}_{d,n}^2(t)\}$$

, and report $\bar{\mathcal{C}}_{n,\alpha}^d = \bar{\mathcal{A}}_{n,\alpha}^d$ when this set is nonempty and $\bar{\mathcal{C}}_{n,\alpha}^d = \{0\}$ otherwise.

Inputs. Z_n ; $\hat{e}_{-k,n}$; d ; α ; ε .

6 Main results

Lemma 1 (lem:identification). *Known supporting result. Under consistency, conditional exchangeability, monotonicity, and positive directional factual denominators, $\tilde{P}_n\{Y(0) = 0 \mid A = 1, Y(1) = 1\} = \Theta_{\text{PN}}(P_n)$ and $\tilde{P}_n\{Y(1) = 1 \mid A = 0, Y(0) = 0\} = \Theta_{\text{PS}}(P_n)$, invariant across admitted full-data extensions with observed margin P_n . Under overlap, $E_{P_n}[\psi_{\Theta_d(P_n),n}^d(O; \eta_n^d)] = 0$.*

Proof. Suppress the row index. Consistency and conditional exchangeability imply, for $a \in \{0, 1\}$,

$$E\{Y(a) \mid A = a, X\} = E\{Y(a) \mid X\} = m_a(X). \quad (1)$$

For PN, positivity of the factual denominator and consistency give

$$\tilde{P}\{A = 1, Y(1) = 1\} = E[e(X)m_1(X)] = E[AY] > 0. \quad (2)$$

Binary monotonicity gives $P\{Y(0) = 0, Y(1) = 1 \mid X\} = m_1(X) - m_0(X)$. Conditional exchangeability of the pair of potential outcomes and A , followed by (2), therefore yields

$$\tilde{P}\{Y(0) = 0 \mid A = 1, Y(1) = 1\} = \frac{E[e(X)\{m_1(X) - m_0(X)\}]}{E[e(X)m_1(X)]} = \Theta_{\text{PN}}(P). \quad (3)$$

For PS,

$$\tilde{P}\{A = 0, Y(0) = 0\} = E[(1 - e(X))\{1 - m_0(X)\}] > 0, \quad (4)$$

and the numerator of $\{Y(1) = 1\}$ inside that event is $E[(1 - e(X))\{m_1(X) - m_0(X)\}]$. Hence

$$\tilde{P}\{Y(1) = 1 \mid A = 0, Y(0) = 0\} = 1 - \frac{E[(1 - e(X))\{1 - m_1(X)\}]}{E[(1 - e(X))\{1 - m_0(X)\}]} = \Theta_{\text{PS}}(P). \quad (5)$$

The right sides of (3)–(5) depend only on the observed margin, proving extension invariance. Finally, iterated expectation and overlap give

$$E[\psi_t^{\text{PN}}(O; \eta)] = E[e(X)\{m_1(X) - m_0(X)\}] - tE[AY]. \quad (6)$$

At $t = \Theta_{\text{PN}}(P)$, `def:functionals` makes the right side zero. Applying the same calculation to $(X, 1 - A, 1 - Y)$, as prescribed by `def:scores`, proves PS centering. $\square$

Justification. This supporting lemma fixes the known causal interpretations of the observed-law PN and PS functionals before the rare-event inference analysis. *Gap.* No identification novelty is claimed; the formulas are the established monotonicity benchmarks. *Consumer.* It interprets externally fixed climate and stroke attribution targets without validating event construction, law selection, or shell membership.

Proposition 1 (prop:causal-localization). On $\mathcal{B}_n^{\text{PN}}$, **prop:pn-causal-localization** gives $m_{0,n}(X) \leq m_{1,n}(X)$ $P_{X,n}$ -almost surely,

$$E[m_{0,n}(X)] \leq E[m_{1,n}(X)] \leq \varepsilon^{-1} q_{\text{PN}}(P_n),$$

and the PN fitted-score remainder is

$$O_{P_n^{\otimes n}} \left(\frac{a_{0,n}}{\sqrt{n}} + a_{e,n} \sqrt{\frac{q_{\text{PN}}(P_n)}{n}} + a_{e,n} a_{0,n} \right),$$

which is $o_{P_n^{\otimes n}}(\sqrt{r_n/n})$ on every schedule in $\mathcal{R}^{\text{PN}}$. On $\mathcal{B}_n^{\text{PS}}$, **prop:ps-causal-localization** gives $1 - m_{1,n}(X) \leq 1 - m_{0,n}(X)$ $P_{X,n}$ -almost surely,

$$E[1 - m_{1,n}(X)] \leq E[1 - m_{0,n}(X)] \leq \varepsilon^{-1} q_{\text{PS}}(P_n),$$

and the PS fitted-score remainder is

$$O_{P_n^{\otimes n}} \left(\frac{a_{1,n}}{\sqrt{n}} + a_{e,n} \sqrt{\frac{q_{\text{PS}}(P_n)}{n}} + a_{e,n} a_{1,n} \right),$$

which is $o_{P_n^{\otimes n}}(\sqrt{r_n/n})$ on every schedule in $\mathcal{R}^{\text{PS}}$.

Proof. There are two cases. If $d = \text{PN}$, every inequality and the displayed remainder are exactly the conclusions of **prop:pn-causal-localization**; in particular that proposition already states the $o_{P_n^{\otimes n}}(\sqrt{r_n/n})$ conclusion on every PN schedule. If $d = \text{PS}$, every inequality and the displayed remainder are exactly the conclusions of **prop:ps-causal-localization**, including its corresponding little-oh conclusion. The identities $b(\text{PN}) = 0$ and $b(\text{PS}) = 1$ match the subscripts in the two displays. These cases exhaust the direction set, and each cited proposition is formulated on its own directional upper class. $\square$

Justification. The corrected proposition localizes PN on $\mathcal{B}_n^{\text{PN}}$ and PS on $\mathcal{B}_n^{\text{PS}}$, avoiding any import of the opposite regression's smoothness. *Gap.* The result is a sufficient remainder calculation for the declared learners, not a universal necessity boundary. *Consumer.* The direction-specific Gaussian upper proofs consume the matching branch.

Lemma 2 (lem:lf-membership). For every LF-compatible primitive tuple, every $d \in \{\text{PN}, \text{PS}\}$, every $(c_G, C_G) \in \mathcal{K}_{\text{LF}}$, and every $(r_n) \in \mathcal{R}^d$ satisfying $r_n \leq \bar{d}$ for all sufficiently large n , the construction in **def:least-favorable** at $r = r_n$, including $t_{\pm} = 1/2 \pm h(nr_n)^{-1/2}$ for bounded sufficiently small h , lies in the exact direction-specific shell $\mathcal{M}_{G,n}^d(r_n; c_G, C_G)$ for all sufficiently large n . In the PN case,

$$(e, m_0, m_1) = \left(\frac{1}{2}, 2(1-t)r, 2r \right),$$

whereas in the PS case,

$$(e, m_0, m_1) = \left(\frac{1}{2}, 1-2r, 1-2(1-t)r \right).$$

In both cases $q_d(P) = r$, $\Theta_d(P) = t$, and $V_d(P)/r \in [3/16, 21/16]$; the two product laws have Kullback–Leibler divergence $O(h^2)$. Thus every such eventually capped direction-specific shell is eventually nonempty.

Proof. Fix an eventually capped schedule and take a sufficiently large row. Since $r = r_n \leq \bar{d} \leq 1/2$, the three masses in **def:least-favorable** are nonnegative and sum to one. The construction has consistency by definition, treatment is independent of $(X, Y(0), Y(1))$, and the only potential-outcome types are $(0, 0), (0, 1), (1, 1)$; hence

exchangeability and monotonicity hold. Its design density is one and $e = 1/2$, so LF compatibility gives the density, overlap, and Hölder requirements. In the PN case the regressions are

$$(e, m_0, m_1) = \left(\frac{1}{2}, 2(1-t)r, 2r\right), \quad (1)$$

and in the transformed PS case they are

$$(e, m_0, m_1) = \left(\frac{1}{2}, 1-2r, 1-2(1-t)r\right). \quad (2)$$

The transformation defining (2) preserves all causal conditions. Direct substitution in `def:functionals` gives, in either direction,

$$q_d(P) = r, \quad \Theta_d(P) = t. \quad (3)$$

For PN put $a = 1 - t$. At the true target, the score is $aY - 2ar$ on the treated arm and $-(Y - 2ar)$ on the control arm. Therefore

$$\frac{V_{\text{PN}}(P)}{r} = a^2(1-2r) + a(1-2ar) = a + a^2(1-4r). \quad (4)$$

Because $a \in [1/4, 3/4]$ and $0 < r \leq 1/2$, the right side lies between $a(1-a)$ and $a + a^2$, hence in $[3/16, 21/16]$. The explicit PS bijection preserves this score law, so (4) and the same bounds hold for PS. Since $(c_G, C_G) \in \mathcal{K}_{\text{LF}}$, $c_G \leq 3/16$, and shell membership follows from (3)–(4).

It remains to check the local pair. By `lem:upper-schedule-consequences`, $nr_n \rightarrow \infty$, so $t_{\pm} = 1/2 \pm h/\sqrt{nr_n}$ eventually lie in $[1/4, 3/4]$ for bounded sufficiently small h . In the PN observed law only the two control-arm outcome cells change with t . Their probability difference in absolute value is

$$\delta_n = (t_+ - t_-)r_n = \frac{2hr_n}{\sqrt{nr_n}}. \quad (5)$$

The smaller rare-cell probability is at least $r_n/4$, and the complementary control-cell probability is at least $1/8$. The elementary inequality $\text{KL}(p, q) \leq \sum_z \{p(z) - q(z)\}^2 / q(z)$ therefore gives

$$\text{KL}(P_{+,n}, P_{-,n}) \leq C \frac{\delta_n^2}{r_n} \leq C \frac{h^2}{n}. \quad (6)$$

The same bound holds in reverse and after the PS bijection. Independence makes product Kullback–Leibler divergence n times (6), hence $O(h^2)$. This proves every claimed membership, information, affinity, and nonemptiness assertion. $\square$

Justification. The casewise statement displays the distinct PN and PS nuisance triples and retains the common target, information, and affinity conclusions. *Gap.* This capped witness alone does not cover schedules above $\bar{d}$. *Consumer.* The capped Gaussian lower bound and bounded-count frontier use this exact-shell certificate.

Theorem 1 (`conj:gaussian-lower`). *Fix $\alpha \in (0, 1/2)$, $d \in \{\text{PN}, \text{PS}\}$, an LF-compatible primitive tuple, $(c_G, C_G) \in \mathcal{K}_{\text{LF}}$, and $(r_n) \in \mathcal{R}^d$ such that $r_n \leq \bar{d}$ for all sufficiently large n . Every measurable $D_n \subseteq [0, 1]$ satisfying*

$$\liminf_n \inf_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} P_n^{\otimes n} \{\Theta_d(P_n) \in D_n\} \geq 1 - \alpha$$

obeys

$$\liminf_n \sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} (nr_n)^{1/2} E_{P_n^{\otimes n}}[\text{diam}(D_n)] \geq c_\alpha > 0$$

Proof. Fix $0 < \eta < (1 - 2\alpha)/4$. By the assumed uniform coverage, for every sufficiently large n and every law in the shell,

$$P_n^{\otimes n} \{\Theta_d(P_n) \in D_n\} \geq 1 - \alpha - \eta. \quad (1)$$

Choose the two laws $P_{+,n}, P_{-,n}$ from `lem:lf-membership` with $t_{\pm} = 1/2 \pm h/\sqrt{nr_n}$. That lemma places both laws in the identical shell and gives $\text{KL}(P_{+,n}^{\otimes n}, P_{-,n}^{\otimes n}) \leq Ch^2$. Pinsker's inequality, which follows from $(u-1)\log u \geq 2(u-1)^2/(u+1)$ integrated against the dominated pair, yields

$$\|P_{+,n}^{\otimes n} - P_{-,n}^{\otimes n}\|_{\text{TV}} \leq C_0 h. \quad (2)$$

Choose $h > 0$ so small that $C_0 h \leq (1-2\alpha)/4$; this choice depends only on α and the fixed primitive constants. By (1)–(2), under $P_{+,n}^{\otimes n}$,

$$\begin{aligned} P_{+,n}^{\otimes n}\{t_+, t_- \in D_n\} &\geq P_{+,n}^{\otimes n}\{t_+ \in D_n\} + P_{+,n}^{\otimes n}\{t_- \in D_n\} - 1 \\ &\geq 1 - 2\alpha - 2\eta - C_0 h =: \kappa_{\alpha} > 0. \end{aligned} \quad (3)$$

On the event in (3), $\text{diam}(D_n) \geq |t_+ - t_-| = 2h/\sqrt{nr_n}$. Hence

$$\sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} \sqrt{nr_n} E_{P_n^{\otimes n}}[\text{diam}(D_n)] \geq 2h\kappa_{\alpha}. \quad (4)$$

Taking the lower limit proves the result with $c_{\alpha} = 2h\kappa_{\alpha} > 0$. This is the standard two-point Le Cam route; the contribution of the preceding membership lemma is that both testing points satisfy the causal assumptions and lie in the same directional shell. $\square$

Justification. The explicit least-favorable pair gives the matching expected-diameter converse inside the identical Gaussian shell. *Gap.* The two-point technique is standard; the causal same-shell witness is result-specific. *Consumer.* It certifies that no uniformly honest directional set improves the effective-count rate on this shell.

Proposition 2 (`prop:regular-reduction`). *Known sanity reduction. If $\inf_n r_n > 0$, then $nr_n \rightarrow \infty$ is automatic and the single retained product-rate condition in `def:admissible-schedules` is equivalent to $a_{e,n}a_{b(d),n} = o(n^{-1/2})$. The proposed directional score formulas agree with TianWu2025 Section 3.1 and Theorem 1 for PN, Section 4.2 and Theorem 3 for PN inference, and Section 7.1–7.2 with Theorems 5–6 for PS, together with the regular variance-inference context of CaiKuroki2005.*

Proof. Put $b = b(d)$, let $(r_n) \in \mathcal{R}^d$, and suppose that $c := \inf_n r_n > 0$. By `def:admissible-schedules`, $0 < r_n \leq 1$, and the retained product-rate condition is

$$\frac{r_n}{na_{e,n}^2 a_{b,n}^2} \longrightarrow \infty. \quad (1)$$

Because every term is positive, (1) is equivalent to

$$\frac{na_{e,n}^2 a_{b,n}^2}{r_n} \longrightarrow 0. \quad (2)$$

If (2) holds, then $r_n \leq 1$ gives

$$0 \leq na_{e,n}^2 a_{b,n}^2 = r_n \frac{na_{e,n}^2 a_{b,n}^2}{r_n} \leq \frac{na_{e,n}^2 a_{b,n}^2}{r_n} \longrightarrow 0. \quad (3)$$

Conversely, if $na_{e,n}^2 a_{b,n}^2 \rightarrow 0$, then $r_n \geq c$ gives

$$0 \leq \frac{na_{e,n}^2 a_{b,n}^2}{r_n} \leq c^{-1} na_{e,n}^2 a_{b,n}^2 \longrightarrow 0. \quad (4)$$

Thus (1) is equivalent to $\sqrt{n} a_{e,n} a_{b(d),n} \rightarrow 0$, or $a_{e,n} a_{b(d),n} = o(n^{-1/2})$. Moreover, $nr_n \geq cn \rightarrow \infty$, so effective-count divergence is automatic.

Suppress the row index for the formula comparison. By iterated expectation and the definitions of the observed regressions,

$$q_{\text{PN}} = E[AY] = E[e(X)m_1(X)], \quad q_{\text{PS}} = E[(1-A)(1-Y)] = E[(1-e(X))(1-m_0(X))]. \quad (5)$$

Write $\beta = \Theta_{\text{PN}}(P)$ and $\gamma = \Theta_{\text{PS}}(P)$. The two formulas in `def:functionals`, together with (5), imply

$$\beta = 1 - \frac{E[e(X)m_0(X)]}{q_{\text{PN}}}, \quad \gamma = \frac{E[(1-e(X))\{m_1(X) - m_0(X)\}]}{q_{\text{PS}}}. \quad (6)$$

For the second equality, subtract the numerator in the defining display for Θ_{PS} from its denominator. These are the identifying ratios recorded in **lem:tian-wu-regular-benchmark**.

For PN, direct expansion of **def:scores** at $t = \beta$ gives

$$\frac{\psi_{\beta}^{\text{PN}}(O; \eta^{\text{PN}})}{q_{\text{PN}}} = \frac{1 - \beta}{q_{\text{PN}}} AY - \frac{e(X)}{q_{\text{PN}}\{1 - e(X)\}}(1 - A)\{Y - m_0(X)\} - \frac{m_0(X)}{q_{\text{PN}}}A. \quad (7)$$

Thus the true PN score divided by its factual denominator is exactly the PN efficient influence function recorded in **lem:tian-wu-regular-benchmark**. For PS, the defining transformation $(A, Y, e, m_0) \mapsto (1 - A, 1 - Y, 1 - e, 1 - m_1)$ gives

$$\psi_t^{\text{PS}}(O; \eta^{\text{PS}}) = (1 - A)\{m_1(X) - Y\} + \frac{1 - e(X)}{e(X)}A\{Y - m_1(X)\} - t(1 - A)(1 - Y). \quad (8)$$

At $t = \gamma$, equation (8) is q_{PS} times the PS efficient influence function in **lem:tian-wu-regular-benchmark**. Indeed, that benchmark influence function can be written as

$$\begin{aligned} \phi_{\text{PS}}(O) &= \frac{1 - e(X)}{q_{\text{PS}}e(X)}A\{Y - m_1(X)\} - \frac{1 - \gamma}{q_{\text{PS}}}(1 - A)\{Y - m_0(X)\} \\ &\quad + \frac{m_0(X) - m_1(X) + \gamma\{1 - m_0(X)\}}{q_{\text{PS}}}(A - 1). \end{aligned} \quad (9)$$

For $A = 1$, the numerator in (9) equals the numerator in (8). For $A = 0$, the numerator of the last two terms in (9) is

$$-(1 - \gamma)\{Y - m_0(X)\} - \{m_0(X) - m_1(X) + \gamma(1 - m_0(X))\} = m_1(X) - Y - \gamma(1 - Y), \quad (10)$$

which again is the corresponding value of (8). Hence $\psi_{\gamma}^{\text{PS}}/q_{\text{PS}} = \phi_{\text{PS}}$. Overlap and the positive factual denominators make all displayed divisions legitimate.

It remains to compare the empirical ratio and its studentizer. For a held-out observation, write the fitted affine score as

$$\psi_{t,n}^d(O_{i,n}; \hat{\eta}_{d,-k(i),n}) = H_{i,d} - tQ_{i,d}, \quad Q_{i,\text{PN}} = A_{i,n}Y_{i,n}, \quad Q_{i,\text{PS}} = (1 - A_{i,n})(1 - Y_{i,n}), \quad (11)$$

where $H_{i,d}$ is the part independent of t . Therefore **def:score-inversion** gives the exact identity

$$\hat{g}_{d,n}(t) = \hat{h}_{d,n} - t\hat{q}_{d,n}, \quad \hat{h}_{d,n} = \frac{1}{n} \sum_{i=1}^n H_{i,d}, \quad \hat{q}_{d,n} = \frac{1}{n} \sum_{i=1}^n Q_{i,d}. \quad (12)$$

On $\{\hat{q}_{d,n} > 0\}$, its unconstrained root is the cross-fitted ratio

$$\tilde{\Theta}_{d,n} = \frac{\hat{h}_{d,n}}{\hat{q}_{d,n}}. \quad (13)$$

At this root, the empirical variance of the estimated influence-function observations $(H_{i,d} - \tilde{\Theta}_{d,n}Q_{i,d})/\hat{q}_{d,n}$ is exactly

$$\frac{\hat{\sigma}_{d,n}^2(\tilde{\Theta}_{d,n})}{\hat{q}_{d,n}^2}, \quad (14)$$

so its standard error is $\hat{\sigma}_{d,n}(\tilde{\Theta}_{d,n})/(\sqrt{n}\hat{q}_{d,n})$. In particular, (14) retains the observation-specific factual indicator $Q_{i,d}$, as required for the empirical influence-function variance of a ratio root. When $\hat{q}_{d,n} = 0$, **def:score-inversion** remains defined without division. The clipping in **def:learners** keeps inverse propensity weights finite and changes none of the algebraic identities above.

Equations (6)–(14), together with **lem:tian-wu-regular-benchmark**, establish agreement of the PN and PS targets, scores, cross-fitted ratio roots, and regular studentization formulas with the cited Tian–Wu sections and theorems. The rate equivalence in (1)–(4) concerns the direction-specific propensity–outcome-regression product appearing among their sufficient conditions; it does not identify that one rate with every condition in their regular theorem. Finally, **lem:cai-kuroki-regular-variance-context** supplies the stated historical regular variance-estimation context and makes no rare-event or honest-diameter claim. $\square$

Justification. For a factual schedule bounded away from zero, the retained rare-event product condition is exactly the direction-specific root- n nuisance-product condition, and the affine PN/PS scores reduce algebraically to the cited regular efficient-influence-function formulas. *Gap.* This is a sanity check against the regular benchmark, not a novelty claim and not an equivalence with every condition in Tian and Wu’s regular theorem. *Consumer.* It tells regular-regime users how the rare-event score inversion, ratio root, and observation-specific factual-indicator studentizer specialize when r_n does not vanish.

Theorem 2 (thm:two-nuisance-learning-boundary-not-necessary). *The two inequalities in `oeq:sharp-learning-boundary` are sufficient for the declared two-nuisance score but are not necessary for first-order cross-fitted score inversion in the causal model. More precisely, fix $d \in \{\text{PN}, \text{PS}\}$, write $b = b(d)$, and choose*

$$\max\{1 - 2s_e, 2s_b, 2s_e + 2s_b - 1\} < \gamma < 1, \quad r_n = \bar{d} \wedge n^{-\gamma}.$$

Then $nr_n \rightarrow \infty$, $nr_n a_{e,n}^2 \rightarrow 0$, but

$$r_n/a_{b,n}^2 \rightarrow 0, \quad r_n/\{na_{e,n}^2 a_{b,n}^2\} \rightarrow 0.$$

On every eventually nonempty propensity-only raw shell $\mathcal{S}_{G,n}^{\partial,d}(r_n; c_G)$, the first-order cross-fitted affine inversion $\bar{C}_{n,\alpha}^d$ nevertheless has uniform asymptotic coverage at least $1 - \alpha$ and uniform expected diameter $O((nr_n)^{-1/2})$. The same conclusions hold on every nonempty sub-shell obtained by intersecting this raw shell with the matching directional upper class $\mathcal{B}_n^d$. For LF-compatible primitives and $0 < c_G \leq 3/16$, those raw shells are nonempty. Formally, $\mathcal{M}_{G,n}^d(r_n; c_G, C_G)$ in the frozen core is indexed only by $(r_n) \in \mathcal{R}^d$, and membership in $\mathcal{R}^d$ already assumes the product inequality and implies the single-regression inequality. Thus necessity cannot be tested inside that restricted indexing domain; on the larger raw shells carrying the same causal and score-information restrictions, the claimed universal necessity is false.

Proof. The schedule definition `def:admissible-schedules` requires

$$\frac{r_n}{na_{e,n}^2 a_{b,n}^2} \rightarrow \infty,$$

and `lem:upper-schedule-consequences` derives $r_n/a_{b,n}^2 \rightarrow \infty$. Hence, within the frozen indexing domain $\mathcal{R}^d$, the proposed necessity conditions are imposed rather than testable.

To test universal necessity, retain the same causal, overlap, and information restrictions but use the raw boundary shells. Since $s_e, s_b < 1/2$ and $s_e + s_b < 1$, the maximum in the theorem statement is strictly less than one, so such a γ exists. The explicit power calculation gives

$$nr_n = n^{1-\gamma} \rightarrow \infty, \quad nr_n a_{e,n}^2 = n^{1-\gamma-2s_e} (\log n)^{2s_e} \rightarrow 0,$$

and

$$\frac{r_n}{a_{b,n}^2} = \frac{n^{2s_b-\gamma}}{(\log n)^{2s_b}} \rightarrow 0, \quad \frac{r_n}{na_{e,n}^2 a_{b,n}^2} = \frac{n^{2s_e+2s_b-1-\gamma}}{(\log n)^{2s_e+2s_b}} \rightarrow 0.$$

The finite initial effect of capping by $\bar{d}$ does not alter any limit. The inference and expected-diameter conclusions now follow directly from `thm:ultra-rare-boundary-score-robustness`. Because uniform coverage and a supremum expected-diameter bound pass to subsets, they also hold after intersecting the raw shell with $\mathcal{B}_n^{\text{PN}}$ or $\mathcal{B}_n^{\text{PS}}$, as appropriate. Its final clause, via the frozen least-favorable witness, gives nonemptiness under LF compatibility and $c_G \leq 3/16$. Thus a first-order cross-fitted score attains the target diameter order while both proposed inequalities fail in the reverse direction, proving that they are not estimator-general necessity conditions. $\square$

Theorem 3 (thm:bounded-count-diameter-lower-bound). *Fix $\alpha \in (0, 1/2)$, $d \in \{\text{PN}, \text{PS}\}$, an LF-compatible primitive tuple, and a deterministic sequence $0 < r_n \leq \bar{d}$ such that $M := \sup_{n \geq 1} nr_n < \infty$. Define*

$$\Delta_{\alpha,M} = \frac{1 - 2\alpha}{4\sqrt{M \vee 1}}.$$

Every measurable random set $D_n \subseteq [0, 1]$ satisfying

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{M}_{B,n}^d(r_n)} P^{\otimes n} \{\Theta_d(P) \in D_n\} \geq 1 - \alpha$$

obeys the full-sequence lower bound

$$\liminf_{n \rightarrow \infty} \sup_{P \in \mathcal{M}_{B,n}^d(r_n)} E_{P^{\otimes n}}[\text{diam}(D_n)] \geq \frac{(1-2\alpha)\Delta_{\alpha,M}}{2} = \frac{(1-2\alpha)^2}{8\sqrt{M \vee 1}} > 0.$$

Consequently the bounded-count question has an affirmative answer, and one may take the subsequence $n_k = k$. The displayed constant is uniform over the fixed shell sequence and makes explicit its dependence on the bounded-count ceiling M .

Proof. Put $\Delta = \Delta_{\alpha,M}$ and $t_- = 1/2 - \Delta/2$, $t_+ = 1/2 + \Delta/2$. Since $0 < 1 - 2\alpha < 1$ and $M \vee 1 \geq 1$, we have $0 < \Delta < 1/4$, so $t_-, t_+ \in [1/4, 3/4]$. For each row n and sign $s \in \{-, +\}$, let $P_{s,n}$ be the direction- d observed law generated by `def:least-favorable` with $r = r_n$ and $t = t_s$. LF compatibility, together with $0 < r_n \leq \bar{d}$, ensures that the uniform design, the constant propensity, and all constant outcome regressions in this construction meet the density, overlap, and declared Hölder-ball restrictions in $\mathcal{B}_n$. The displayed potential-outcome type probabilities are nonnegative and sum to one by $r_n \leq \bar{d} \leq 1/2$; they satisfy consistency, conditional exchangeability, and monotonicity by construction. In the PN construction,

$$(e, m_0, m_1) = \left(\frac{1}{2}, 2(1 - t_s)r_n, 2r_n \right),$$

so $q_{\text{PN}}(P_{s,n}) = r_n$ and direct substitution in `def:functionals` gives $\Theta_{\text{PN}}(P_{s,n}) = t_s$. In the PS construction,

$$(e, m_0, m_1) = \left(\frac{1}{2}, 1 - 2r_n, 1 - 2(1 - t_s)r_n \right),$$

and the same substitution gives $q_{\text{PS}}(P_{s,n}) = r_n$ and $\Theta_{\text{PS}}(P_{s,n}) = t_s$. Hence, in either direction,

$$P_{s,n} \in \mathcal{M}_{B,n}^d(r_n), \quad \Theta_d(P_{s,n}) = t_s. \quad (1)$$

We next control the testing affinity of the two laws. For PN, after integrating out the common uniform covariate, the four probabilities of $(A, Y) = (1, 1), (1, 0), (0, 1), (0, 0)$ under parameter t are

$$r_n, \quad \frac{1}{2} - r_n, \quad (1 - t)r_n, \quad \frac{1}{2} - (1 - t)r_n. \quad (2)$$

The PS construction is the bijective transformation $(A, Y) \mapsto (1 - A, 1 - Y)$, so it has exactly the same divergence between two values of t . Only the last two masses in (2) depend on t . Taking $P_{-,n}$ as the first argument and $P_{+,n}$ as the second, the elementary inequality $\log u \leq u - 1$ gives $\text{KL}(P_{-,n}, P_{+,n}) \leq \chi^2(P_{-,n}, P_{+,n})$. Moreover, because $t_+ \in [1/4, 3/4]$ and $r_n \leq 1/2$,

$$\begin{aligned} \chi^2(P_{-,n}, P_{+,n}) &= \frac{\Delta^2 r_n^2}{(1 - t_+)r_n} + \frac{\Delta^2 r_n^2}{1/2 - (1 - t_+)r_n} \\ &\leq 4\Delta^2 r_n + 8\Delta^2 r_n^2 \leq 8\Delta^2 r_n. \end{aligned} \quad (3)$$

The common covariate factor contributes zero divergence. Row-wise independent sampling and additivity of product Kullback–Leibler divergence therefore yield

$$\text{KL}(P_{-,n}^{\otimes n}, P_{+,n}^{\otimes n}) \leq 8nr_n\Delta^2. \quad (4)$$

By `lem:elementary-total-variation-from-kl`,

$$\|P_{-,n}^{\otimes n} - P_{+,n}^{\otimes n}\|_{\text{TV}} \leq 2\Delta\sqrt{nr_n} \leq 2\Delta\sqrt{M} \leq \frac{1 - 2\alpha}{2} =: \rho. \quad (5)$$

Let $E_{s,n} = \{t_s \in D_n\}$. By the assumed uniform coverage and membership (1), for every $\xi > 0$ and all sufficiently large n ,

$$P_{s,n}^{\otimes n}(E_{s,n}) \geq 1 - \alpha - \xi, \quad s \in \{-, +\}. \quad (6)$$

Total-variation transfer using (5), followed by the union bound, gives

$$\begin{aligned} P_{-,n}^{\otimes n}(E_{-,n} \cap E_{+,n}) &\geq P_{-,n}^{\otimes n}(E_{-,n}) + P_{-,n}^{\otimes n}(E_{+,n}) - 1 \\ &\geq P_{-,n}^{\otimes n}(E_{-,n}) + P_{+,n}^{\otimes n}(E_{+,n}) - 1 - \rho \\ &\geq 1 - 2\alpha - 2\xi - \rho. \end{aligned} \tag{7}$$

On $E_{-,n} \cap E_{+,n}$, the set D_n contains two points at distance $t_+ - t_- = \Delta$, and hence $\text{diam}(D_n) \geq \Delta$. Since $P_{-,n}$ is one of the laws over which the supremum is taken, (7) implies

$$\sup_{P \in \mathcal{M}_{B,n}^d(r_n)} E_{P^{\otimes n}}[\text{diam}(D_n)] \geq \Delta(1 - 2\alpha - 2\xi - \rho)$$

for all sufficiently large n . Taking the lower limit, then letting $\xi \downarrow 0$, and using $\rho \leq (1 - 2\alpha)/2$, proves

$$\liminf_{n \rightarrow \infty} \sup_{P \in \mathcal{M}_{B,n}^d(r_n)} E_{P^{\otimes n}}[\text{diam}(D_n)] \geq \frac{(1 - 2\alpha)\Delta}{2} = \frac{(1 - 2\alpha)^2}{8\sqrt{M} \vee 1}.$$

This is a full-sequence assertion, so the requested subsequence may be chosen as $n_k = k$. $\square$

Lemma 3 (lem:cheng-mao-pearl-li-2026-comparator). *Cheng, Mao, Pearl, and Li study generally partially identified probability-of-causation bound endpoints in finite discrete models. Their endpoints are finite minima or maxima of explicit linear or ratio-type functions of finitely many experimental and observational probabilities. They use an ordinary multivariate delta method at smooth unique-active endpoints, a directional numerical delta method at nonsmooth endpoints, and margin-of-error calculations for sample-size choice.*

Justification. The cited leaf fixes the finite-discrete partial-identification comparator's exact scope and methods. *Gap.* Its positive-denominator endpoint delta method does not cover a point-identified score whose variance vanishes along a triangular array. *Consumer.* It disciplines the non-subsumption claim in the Gaussian-frontier framing.

Lemma 4 (lem:cai-low-two-point-confidence-length-route). *For a normal observation $X \sim N(\theta, \sigma^2)$ and a two-point parameter set $\{\theta_0, \theta_1\}$, Cai and Low's Proposition 1 gives the standard honest-confidence-length lower bound*

$$E_{\theta_i} L(CI) \geq (\theta_1 - \theta_0) \left[1 - \alpha - \Phi \left\{ \frac{\theta_1 - \theta_0}{\sigma} - z_\alpha \right\} \right]_+$$

for at least one endpoint. It is the published two-point route to a confidence-set diameter converse; it does not provide the causal least-favorable shell or the matching PN/PS upper theorem.

Lemma 5 (lem:derumigny-weak-ratio-comparator). *Derumigny, Girard, and Guyonvarch study ratios of expectations under sequences in which the denominator expectation may decrease to zero. They generalize the delta method, analyze percentile-bootstrap consistency, and prove a partial impossibility for nonasymptotic uniform ratio confidence intervals. Their generic ratio theory does not supply causal PN/PS nuisance localization or a matched expected-diameter frontier.*

Lemma 6 (lem:smucler-lanni-masip-score-inversion-comparator). *Smucler, Lanni, and Masip analyze a cross-fitted score-test-inversion confidence set for LATE under arbitrarily weak instruments. They characterize six possible set shapes and prove fixed-law equivalence, up to order $1/n$, with the doubly robust Wald interval; their work gives neither an expected-diameter rate over drifting shells nor a minimax diameter converse.*

Lemma 7 (lem:ma-identification-robust-late-comparator). *Ma constructs identification-robust LATE inference with high-dimensional covariates by inverting a cross-fitted, Neyman-orthogonal test. The procedure controls asymptotic size without restricting identification strength. It is a weak-instrument size theorem, not an expected-diameter frontier for a bounded causal ratio whose factual-event probability determines a drifting score scale.*

Lemma 8 (lem:generic). *Under consistency, conditional exchangeability, and overlap, for each $a \in \{0, 1\}$, $m_{a,n}(x) = E_{\tilde{P}_n}[Y(a) \mid X = x]$ for $P_{X,n}$ -almost every x . Under monotonicity, $m_{0,n}(x) \leq m_{1,n}(x)$ for $P_{X,n}$ -almost every x .*

Proof. Overlap makes both treatment values have positive conditional probability for $P_{X,n}$ -almost every x . Thus consistency and conditional exchangeability imply, for $a \in \{0, 1\}$,

$$m_{a,n}(x) = E[Y \mid A = a, X = x] = E[Y(a) \mid A = a, X = x] = E[Y(a) \mid X = x]. \quad (1)$$

Monotonicity states $Y(0) \leq Y(1)$ almost surely. Taking conditional expectations given $X = x$ in that inequality and using (1) gives $m_{0,n}(x) \leq m_{1,n}(x)$ for $P_{X,n}$ -almost every x . $\square$

Lemma 9 (lem:bounded-iid-normal-approximation). *Let $\mathcal{Q}_n$ be a triangular array of classes of laws. Under each $Q \in \mathcal{Q}_n$, let $U_{1,n}, \dots, U_{n,n}$ be independent and identically distributed with $E_Q U_{i,n} = 0$, $E_Q U_{i,n}^2 = v_n(Q) > 0$, and $|U_{i,n}| \leq B$ almost surely for a common finite B . If $\inf_{Q \in \mathcal{Q}_n} n v_n(Q) \rightarrow \infty$, then, with Φ the standard-normal distribution function,*

$$\sup_{Q \in \mathcal{Q}_n} \sup_{x \in \mathbb{R}} \left| Q \left\{ \frac{\sum_{i=1}^n U_{i,n}}{\sqrt{n v_n(Q)}} \leq x \right\} - \Phi(x) \right| \rightarrow 0.$$

Proof. Fix $Q \in \mathcal{Q}_n$, write $v = v_n(Q)$, and let $X_i = U_{i,n}/\sqrt{nv}$. Let $G_1, \dots, G_n$ be independent $N(0, 1/n)$ variables. The pairs (X_i, G_i) have matching first two moments, and

$$\sum_{i=1}^n E|X_i|^3 = \frac{E|U_{1,n}|^3}{\sqrt{n} v^{3/2}} \leq \frac{B}{\sqrt{nv}}, \quad \sum_{i=1}^n E|G_i|^3 = \frac{E|N(0, 1)|^3}{\sqrt{n}}. \quad (1)$$

For $0 < \delta < 1$, choose a thrice continuously differentiable cutoff $f_{x,\delta}$ which is one on $(-\infty, x]$, zero on $[x + \delta, \infty)$, takes values in $[0, 1]$, and satisfies $\|f_{x,\delta}^{(3)}\|_\infty \leq C\delta^{-3}$. Replacing the X_i 's successively by the G_i 's and Taylor-expanding through order two cancels the first two moments. The integral remainder and (1) give

$$\left| E f_{x,\delta} \left(\sum_i X_i \right) - E f_{x,\delta} \left(\sum_i G_i \right) \right| \leq C\delta^{-3} \left\{ \frac{B}{\sqrt{nv}} + \frac{1}{\sqrt{n}} \right\}. \quad (2)$$

The Gaussian sum is standard normal and its density is bounded. Cutoff inequalities at x and $x - \delta$ therefore imply

$$\sup_x \left| Q \left\{ \sum_i X_i \leq x \right\} - \Phi(x) \right| \leq C\delta + C\delta^{-3} \left\{ \frac{B}{\sqrt{nv}} + \frac{1}{\sqrt{n}} \right\}. \quad (3)$$

Put $\rho_n = \sup_{Q \in \mathcal{Q}_n} \{B/\sqrt{nv_n(Q)} + n^{-1/2}\}$. The hypothesis gives $\rho_n \rightarrow 0$. Taking $\delta = \rho_n^{1/4}$ in (3) proves the stated uniform convergence. $\square$

Lemma 10 (lem:upper-schedule-consequences). *For either direction d , every $(r_n) \in \mathcal{R}^d$ satisfies $0 < r_n \leq 1$, $a_{b(d),n}^2/r_n \rightarrow 0$, and $nr_n \rightarrow \infty$.*

Proof. Write $b = b(d)$. The schedule definition gives

$$\frac{r_n}{na_{e,n}^2 a_{b,n}^2} \rightarrow \infty. \quad (1)$$

Since $s_e < 1/2$, $na_{e,n}^2 \rightarrow \infty$. Multiplying this fact by (1) gives $r_n/a_{b,n}^2 \rightarrow \infty$, equivalently $a_{b,n}^2/r_n \rightarrow 0$. Moreover $s_e + s_b < 1$, so $n^2 a_{e,n}^2 a_{b,n}^2 \rightarrow \infty$. Multiplication by (1) gives $nr_n \rightarrow \infty$. Finally $0 < r_n \leq 1$ is part of `def:admissible-schedules`. $\square$

Lemma 11 (lem:pn-structural-score-variance-upper-bound). *For every $P \in \mathcal{B}_n^{\text{PN}}$ with $q_{\text{PN}}(P) > 0$,*

$$V_{\text{PN}}(P) \leq \varepsilon^{-1} q_{\text{PN}}(P).$$

The coefficient ε^{-1} is sharp under the binary monotone and overlap restrictions.

Proof. Put $\pi = e$, $\mu_a = m_a$, $r = E[\pi(X)\mu_1(X)] > 0$, $t = \Theta_{\text{PN}}(P)$, and $a = 1 - t$. By **lem:generic**, $0 \leq \mu_0 \leq \mu_1 \leq 1$ almost surely, while **def:functionals** gives $E[\pi(X)\mu_0(X)] = ar$. Direct substitution in **def:scores** gives, at the true value,

$$\psi_t^{\text{PN}}(O; \eta) = \begin{cases} aY - \mu_0(X), & A = 1, \\ -\pi(X)\{1 - \pi(X)\}^{-1}\{Y - \mu_0(X)\}, & A = 0. \end{cases} \quad (1)$$

The score is centered by **lem:identification**, so its variance is its second moment. The treated- and control-arm contributions are

$$T = E[\pi\{a^2\mu_1 - 2a\mu_1\mu_0 + \mu_0^2\}], \quad C = E\left[\frac{\pi^2}{1 - \pi}\mu_0(1 - \mu_0)\right]. \quad (2)$$

If $a \geq 1/2$, then $\mu_0^2 - 2a\mu_1\mu_0 \leq (1 - 2a)\mu_1\mu_0 \leq 0$, whence $T \leq a^2r \leq r$. If $a < 1/2$, the same first inequality and $\mu_1\mu_0 \leq \mu_0$ give

$$T \leq a^2r + (1 - 2a)E[\pi\mu_0] = \{a^2 + a(1 - 2a)\}r = a(1 - a)r \leq r. \quad (3)$$

Overlap gives $\pi/(1 - \pi) \leq \varepsilon^{-1} - 1$; hence

$$C \leq (\varepsilon^{-1} - 1)E[\pi\mu_0] \leq (\varepsilon^{-1} - 1)r. \quad (4)$$

Equations (2)–(4) prove $V_{\text{PN}}(P) \leq \varepsilon^{-1}r$.

For sharpness, take $\pi \equiv 1 - \varepsilon$ and $Y(0) = Y(1) \sim \text{Bernoulli}(u)$, independent of treatment and covariates. This law is binary, monotone, and satisfies overlap. Here $q_{\text{PN}} = (1 - \varepsilon)u$, and direct evaluation of (1) gives

$$\frac{V_{\text{PN}}}{q_{\text{PN}}} = \frac{1 - u}{\varepsilon} \longrightarrow \varepsilon^{-1} \quad \text{as } u \downarrow 0.$$

Thus no smaller universal coefficient is possible under the stated binary monotone and overlap restrictions. $\square$

Lemma 12 (**lem:ps-structural-score-variance-upper-bound**). *For every $P \in \mathcal{B}_n^{\text{PS}}$ with $q_{\text{PS}}(P) > 0$,*

$$V_{\text{PS}}(P) \leq \varepsilon^{-1}q_{\text{PS}}(P).$$

The coefficient ε^{-1} is sharp under the binary monotone and overlap restrictions.

Proof. Set $A^\dagger = 1 - A$, $Y^\dagger = 1 - Y$, $\pi = 1 - e$, $\mu_0 = 1 - m_1$, and $\mu_1 = 1 - m_0$. Then $0 \leq \mu_0 \leq \mu_1 \leq 1$, overlap is unchanged, and **def:scores** makes the PS score the PN score for $(X, A^\dagger, Y^\dagger)$. With $r = E(\pi\mu_1)$, $t = \Theta_{\text{PS}}(P)$, and $a = 1 - t$, one has $E(\pi\mu_0) = ar$. The PN armwise calculation gives $T \leq r$ and $C \leq (\varepsilon^{-1} - 1)r$, hence $V_{\text{PS}} \leq \varepsilon^{-1}q_{\text{PS}}$. For sharpness take $e = \varepsilon$, $m_0 = m_1 = 1 - u$, and a common potential outcome independent of treatment and covariates. The transformed model has $\pi = 1 - \varepsilon$, $\mu_0 = \mu_1 = u$, and $V_{\text{PS}}/q_{\text{PS}} = (1 - u)/\varepsilon \rightarrow \varepsilon^{-1}$. $\square$

Lemma 13 (**lem:bounded-score-studentization**). *Let $\mathcal{Q}_n$ be a triangular array of law collections. Under every $Q \in \mathcal{Q}_n$, let $U_{1,n}, \dots, U_{n,n}$ be independent and identically distributed with $E_Q U_{i,n} = 0$, $E_Q U_{i,n}^2 = \sigma_n^2(Q) > 0$, and $|U_{i,n}| \leq B$ almost surely for a common finite B . Let $S_n = n^{-1} \sum_{i=1}^n U_{i,n} + R_n$, and let $\hat{\sigma}_n^2 \geq 0$. Suppose that, for every $\delta > 0$,*

$$\inf_{Q \in \mathcal{Q}_n} \sqrt{n} \sigma_n(Q) \longrightarrow \infty, \quad \sup_{Q \in \mathcal{Q}_n} Q \left\{ |R_n| > \delta \frac{\sigma_n(Q)}{\sqrt{n}} \right\} \longrightarrow 0,$$

$$\sup_{Q \in \mathcal{Q}_n} Q \left\{ \left| \frac{\hat{\sigma}_n^2}{\sigma_n^2(Q)} - 1 \right| > \delta \right\} \longrightarrow 0.$$

Define $T_n = \sqrt{n} S_n / \hat{\sigma}_n$ on $\{\hat{\sigma}_n > 0\}$, with any fixed value on its complement. Then

$$\sup_{Q \in \mathcal{Q}_n} \sup_{x \in \mathbb{R}} |Q\{T_n \leq x\} - \Phi(x)| \longrightarrow 0.$$

Consequently, for $z = z_{1-\alpha/2}$,

$$\sup_{Q \in \mathcal{Q}_n} |Q\{|T_n| \leq z\} - (1 - \alpha)| \longrightarrow 0.$$

Proof. Write $\bar{U}_n = n^{-1} \sum_i U_{i,n}$. The first hypothesis and `lem:bounded-iid-normal-approximation` give

$$\sup_Q \sup_x \left| Q \left\{ \frac{\sqrt{n} \bar{U}_n}{\sigma_n(Q)} \leq x \right\} - \Phi(x) \right| \rightarrow 0. \quad (1)$$

Put $A_n = \sqrt{n} \bar{U}_n / \sigma_n(Q)$, $B_n = \sqrt{n} R_n / \sigma_n(Q)$, and $C_n = \hat{\sigma}_n / \sigma_n(Q)$. Uniformly, $B_n = o_Q(1)$, while

$$|C_n - 1| = \frac{|C_n^2 - 1|}{C_n + 1} \leq |C_n^2 - 1| = o_Q(1). \quad (2)$$

In particular $Q(C_n = 0) \rightarrow 0$ uniformly. On $C_n > 0$, $T_n = (A_n + B_n)/C_n$. Fix $M > 0$ and $0 < \delta < 1/2$. On $\{|A_n| \leq M, |B_n| \leq \delta, |C_n - 1| \leq \delta\}$, one has $|T_n - A_n| \leq 2\delta(1 + M)$. The resulting distribution-function sandwich, (1), and the uniform continuity of Φ show that the limsup Kolmogorov distance is at most $P(|Z| > M) + 2\delta(1 + M)/\sqrt{2\pi}$. Letting first $\delta \downarrow 0$ and then $M \uparrow \infty$ proves uniform normality. Evaluation at the two continuity points $\pm z_{1-\alpha/2}$ proves the final assertion. $\square$

Lemma 14 (`lem:control-spline-learning-tails`). *Under row-wise sampling, overlap, support-wide bounded design density, the control-regression Hölder condition, and the PN learner, for every fixed $M > 0$ there is $C_M < \infty$ such that, uniformly over the fixed balanced folds and all sufficiently large n ,*

$$P_n^{\otimes n} \left\{ \max_{1 \leq k \leq K} \frac{\|\hat{m}_{0,-k,n} - m_{0,n}\|_{2,P_n}}{a_{0,n}} > C_M \right\} \leq C_M n^{-M}.$$

Proof. Fix a fold and write $B = B_{0,n}$, $J = J_{0,n}$, and $K_0 = K_{0,n}$. Overlap and bounded design density imply

$$cJ^{-1}\|v\|_2^2 \leq E[(1 - A)\{B(X)^\top v\}^2] \leq CJ^{-1}\|v\|_2^2. \quad (1)$$

Hölder spline approximation gives θ° with $\|B^\top \theta^\circ - m_{0,n}\|_\infty \leq CK_0^{-\beta_0}$. Applying the bounded exponential-series inequality to the locally overlapping Gram and normal-equation coordinates, followed by a union bound, gives deviation order $C_M \{\sqrt{\log n/(nJ)} + \log n/n\}$ outside probability $C_M n^{-M}$. The fixed-range inverse argument used for the propensity Gram and a Neumann perturbation give empirical inverse row-sum norm $O(J)$. The ridge normal equation then yields

$$\|B^\top \hat{\theta}_0 - m_{0,n}\|_{2,P_n} \leq C_M \left\{ K_0^{-\beta_0} + \sqrt{\frac{J \log n}{n}} + \frac{J \log n}{n} + \frac{J}{n^2} \right\} \leq C_M a_{0,n}. \quad (2)$$

Projection onto $[0, 1]$ is contractive toward the true regression. A union bound over the fixed folds proves the claim. No property of $m_{1,n}$ is used. $\square$

Lemma 15 (`lem:treated-spline-learning-tails`). *Under row-wise sampling, overlap, support-wide bounded design density, the treated-regression Hölder condition, and the PS learner, for every fixed $M > 0$ there is $C_M < \infty$ such that, uniformly over the fixed balanced folds and all sufficiently large n ,*

$$P_n^{\otimes n} \left\{ \max_{1 \leq k \leq K} \frac{\|\hat{m}_{1,-k,n} - m_{1,n}\|_{2,P_n}}{a_{1,n}} > C_M \right\} \leq C_M n^{-M}.$$

Proof. Fix a fold and write $B = B_{1,n}$, $J = J_{1,n}$, and $K_1 = K_{1,n}$. Overlap and bounded design density give

$$cJ^{-1}\|v\|_2^2 \leq E[A\{B(X)^\top v\}^2] \leq CJ^{-1}\|v\|_2^2. \quad (1)$$

Hölder spline approximation, bounded-coordinate exponential concentration, and local Gram sparsity give, outside probability $C_M n^{-M}$, a coefficient error bounded by

$$C_M \left\{ K_1^{-\beta_1} + \sqrt{\frac{J \log n}{n}} + \frac{J \log n}{n} + \frac{J}{n^2} \right\}. \quad (2)$$

Indeed, the population inverse row-sum norm is $O(J)$ by fixed-range polynomial approximation of the inverse, the empirical inverse has the same order by a Neumann perturbation, and the ridge normal equation gives (2). With $J \asymp K_1^p$, the declared tuning makes (2) at most $C_M a_{1,n}$ in $L_2(P_n)$. Projection is contractive and a fixed-fold union bound completes the proof. No property of $m_{0,n}$ is used. $\square$

Lemma 16 (lem:boundary-shell-nonempty-witness). *Fix an LF-compatible primitive tuple, $d \in \{\text{PN}, \text{PS}\}$, $0 < c_G \leq 3/16$, $0 < r \leq \bar{d}$, and $t \in [1/4, 3/4]$. The observed law induced by **def:least-favorable** in direction d belongs to $\mathcal{B}_n^\partial$, has $q_d(P) = r$, $\Theta_d(P) = t$, and satisfies $V_d(P) \geq (3/16)r$. Consequently $\mathcal{S}_{G,n}^{\partial,d}(r; c_G)$ is nonempty for every row n .*

Proof. The probability and causal checks in **def:least-favorable** give consistency, exchangeability, monotonicity, overlap, and uniform design density. LF compatibility places the constant propensity $e = 1/2$ in its Hölder ball. Since $\mathcal{B}_n^\partial$ imposes no outcome-regression smoothness, the PN law and its explicit PS transform therefore belong to $\mathcal{B}_n^\partial$. Direct substitution gives $q_d(P) = r$ and $\Theta_d(P) = t$. The score calculation in the witness gives

$$\frac{V_d(P)}{r} = a + a^2(1 - 4r), \quad a = 1 - t \in [1/4, 3/4]. \quad (1)$$

For $0 < r \leq 1/2$, the right side is at least $a(1 - a) \geq 3/16$. Thus $V_d(P) \geq (3/16)r \geq c_G r$, so the law belongs to $\mathcal{S}_{G,n}^{\partial,d}(r; c_G)$. The construction is independent of the row, proving nonemptiness for every n . $\square$

Lemma 17 (lem:elementary-total-variation-from-kl). *For probability laws $P \ll Q$,*

$$\|P - Q\|_{\text{TV}} \leq \sqrt{\text{KL}(P, Q)/2}.$$

Proof. Let $L = dP/dQ$ and set $A = \{L \geq 1\}$. Then

$$\|P - Q\|_{\text{TV}} = P(A) - Q(A). \quad (1)$$

Write $p = P(A)$ and $q = Q(A)$. Conditional Jensen inequality for the convex map $u \mapsto u \log u$ gives

$$\int_A L \log L dQ \geq p \log(p/q), \quad \int_{A^c} L \log L dQ \geq (1 - p) \log\{(1 - p)/(1 - q)\}. \quad (2)$$

The boundary cases follow by the usual convention $0 \log 0 = 0$ and by continuity. Hence

$$\text{KL}(P, Q) \geq k(p, q) := p \log(p/q) + (1 - p) \log\{(1 - p)/(1 - q)\}. \quad (3)$$

For fixed $q \in (0, 1)$, one has $k(q, q) = 0$, $\partial_p k(q, q) = 0$, and

$$\partial_p^2 k(p, q) = \frac{1}{p(1 - p)} \geq 4. \quad (4)$$

Taylor's formula with integral remainder therefore yields $k(p, q) \geq 2(p - q)^2$; continuity gives the same conclusion at the boundary. Combining this inequality with (1)–(3) gives

$$\|P - Q\|_{\text{TV}} \leq \sqrt{\text{KL}(P, Q)/2},$$

as claimed. $\square$

Lemma 18 (lem:uniform-propensity-spline-learning-tails). *Under row-wise sampling, overlap, the support-wide bounded design-density condition, the propensity Hölder condition, and the ridge-regularized clipped propensity learner, for every fixed $M > 0$ there is $C_M < \infty$ such that, uniformly over the fixed balanced folds and all sufficiently large n ,*

$$P_n^{\otimes n} \left\{ \max_{1 \leq k \leq K} \frac{\|\widehat{e}_{-k,n} - e_n\|_\infty}{a_{e,n}} > C_M \right\} \leq C_M n^{-M}.$$

The same assertion holds with the supremum norm replaced by $L_2(P_n)$. No smoothness condition on either outcome regression is required.

Proof. Fix a fold and suppress its index. Its balanced training size N satisfies $N \asymp n$. Write $\beta = \beta_e$, $\kappa = K_{e,n}$, $J = J_{e,n}$, $\lambda = n^{-2}$, and $B(x) = B_{e,n}(x)$. All constants below are uniform over the row laws and fixed folds.

Put $\ell_n = \log\{e_n/(1-e_n)\}$. Overlap and the Hölder composition inequality give $\|\ell_n\|_{\mathcal{H}^\beta} \leq L_\ell$, with L_ℓ depending only on $\varepsilon, \beta, L_e, p$. Degree- r_e tensor-spline quasi-interpolation supplies $\theta_n^\circ \in \mathbb{R}^J$ satisfying

$$\|B^\top \theta_n^\circ - \ell_n\|_\infty \leq C\kappa^{-\beta}, \quad \|\theta_n^\circ\|_\infty \leq \log\{(1-\varepsilon)/\varepsilon\} + C_{\text{QI}} = B_e - 1. \quad (1)$$

Thus the approximant has fixed distance at least one from the boundary of the coefficient cube in **def:learners**.

Nonnegativity, partition of unity, local support, quasi-uniform knots, and $c_X \leq f_{X,n} \leq C_X$ imply the spline stability inequalities

$$cJ^{-1}\|v\|_2^2 \leq v^\top G_n v \leq CJ^{-1}\|v\|_2^2, \quad G_n = E[B(X)B(X)^\top], \quad (2)$$

and $EB_j(X) \leq C/J$. The overlap graph has bounded degree. Applying the elementary bounded exponential-moment inequality coordinatewise to the nonzero Gram entries and to $B_j(X)\{A - e_n(X)\}$, then taking a union bound, gives an event $\mathcal{E}_{M,n}$ with

$$\sup_{P_n} P_n^{\otimes n}(\mathcal{E}_{M,n}^c) \leq C_M n^{-M} \quad (3)$$

on which, for $u_{M,n} = C_M \{\sqrt{\log n/(nJ)} + \log n/n\}$,

$$\max_{j,l} |(G_{N,n} - G_n)_{jl}| \mathbf{1}\{B_j B_l \not\equiv 0\} \vee \max_j |\mathbb{P}_N[B_j(A - e_n)]| \leq u_{M,n}, \quad \max_j \mathbb{P}_N B_j \leq C_M J^{-1}. \quad (4)$$

Here $G_{N,n} = \mathbb{P}_N[BB^\top]$. Since $Ju_{M,n} = o(1)$, bounded graph degree and (2)–(4) imply

$$cJ^{-1}I_J \preceq G_{N,n} \preceq CJ^{-1}I_J. \quad (5)$$

Let L_N be the ridge logistic objective, let $p_n^\circ(x) = \{1 + \exp(-B(x)^\top \theta_n^\circ)\}^{-1}$, and set $s_N = \nabla L_N(\theta_n^\circ)$. The sigmoid is $1/4$ -Lipschitz, so (1) and the local-mass bound in (4) give

$$\|s_N\|_\infty \leq C_M \left\{ \sqrt{\frac{\log n}{nJ}} + \frac{\log n}{n} + \frac{\kappa^{-\beta}}{J} + n^{-2} \right\}. \quad (6)$$

Because $J \asymp \kappa^p$ and $\kappa \asymp (n/\log n)^{1/(2\beta+p)}$,

$$J\|s_N\|_\infty \leq C_M \left\{ \kappa^{-\beta} + \sqrt{\frac{J \log n}{n}} + \frac{J \log n}{n} + \frac{J}{n^2} \right\} \leq C_M a_{e,n}. \quad (7)$$

We now justify the inverse bound and interiority. For $\|\delta\|_\infty \leq R_M a_{e,n}$, define

$$\overline{H}_N(\delta) = \int_0^1 \nabla^2 L_N(\theta_n^\circ + u\delta) du. \quad (8)$$

For large n , $R_M a_{e,n} < 1/2$, and partition of unity plus (1) ensures that every logit on the path in (8) has absolute value below B_e . Hence every logistic weight is at least $w_\star = \exp(B_e)/\{1 + \exp(B_e)\}^2 > 0$, and

$$w_\star G_{N,n} + 2\lambda I_J \preceq \overline{H}_N(\delta) \preceq \frac{1}{4} G_{N,n} + 2\lambda I_J. \quad (9)$$

Thus the spectrum of $J\overline{H}_N(\delta)$ lies in one compact subinterval of $(0, \infty)$. Moreover this matrix has the fixed-range tensor-lattice sparsity graph. Uniform polynomial approximation of x^{-1} on that spectral interval shows that its inverse entries decay geometrically in graph distance. Since the number of indices at distance s is $O((s+1)^{p-1})$, summation gives

$$\sup_{\|\delta\|_\infty \leq R_M a_{e,n}} \|\overline{H}_N(\delta)^{-1}\|_{\infty \rightarrow \infty} \leq CJ. \quad (10)$$

This derives the required infinity-norm inverse bound from the density lower bound and sparse inverse decay, not merely from an eigenvalue comparison.

Define $T_N(\delta) = -\overline{H}_N(\delta)^{-1}s_N$. By (7)–(10), choosing R_M large makes T_N map the cube $\{\|\delta\|_\infty \leq R_M a_{e,n}\}$ into itself. The bounded derivative of the logistic weight, partition of unity, and (4) give

$$\|\overline{H}_N(\delta_1) - \overline{H}_N(\delta_2)\|_{\infty \rightarrow \infty} \leq CJ^{-1} \|\delta_1 - \delta_2\|_\infty$$

. The inverse identity, (7), and (10) therefore make the Lipschitz constant of T_N at most $Ca_{e,n} < 1/2$. Its iterates from zero converge to a fixed point δ_N^* , and the fundamental theorem of calculus yields

$$abla L_N(\theta_n^\circ + \delta_N^*) = s_N + \overline{H}_N(\delta_N^*)\delta_N^* = 0. \quad (11)$$

The ridge makes the unrestricted objective strictly convex and coercive, so this is its unique minimizer. By (1) and $\|\delta_N^*\|_\infty < 1/2$, it lies strictly inside the coefficient cube and is therefore also the constrained minimizer in `def:learners`. This establishes interiority before the score equation is used. Existence, uniqueness, and measurability on every sample follow independently from compact strict convexity and the deterministic projected-gradient construction in that definition.

Finally, nonnegativity and partition of unity, (1), and the sigmoid Lipschitz bound give on $\mathcal{E}_{M,n}$

$$\left\| \frac{\exp(B^\top \hat{\theta}_e)}{1 + \exp(B^\top \hat{\theta}_e)} - e_n \right\|_\infty \leq \frac{1}{4} \|\delta_N^*\|_\infty + C\kappa^{-\beta} \leq C_M a_{e,n}. \quad (12)$$

Projection onto $[\varepsilon/2, 1 - \varepsilon/2]$ is contractive toward e_n , so (12) holds for $\hat{e}_{-k,n}$. Combine (3) and (12), enlarge C_M , and take a union bound over fixed K . The $L_2(P_n)$ conclusion follows from $\|f\|_{2,P_n} \leq \|f\|_\infty$. No outcome variable entered the proof. $\square$

Justification. This is the sole canonical propensity-rate lemma after constraining the fitted logits at the optimization stage. *Gap.* The repaired proof derives the quasi-interpolant’s feasibility margin, the uniform weight floor, and the $O(J_{e,n})$ inverse bound using the design-density lower bound, with explicit constrained-boundary handling. *Consumer.* All score-remainder, studentizer, empirical-envelope, and propensity-only boundary results use this single migrated dependency.

Lemma 19 (lem:pn-score-learning-remainder). *On the PN upper class, the PN learner satisfies $\|\hat{e} - e_n\|_\infty \vee \|\hat{e} - e_n\|_{2,P_n} = O_P(a_{e,n})$, $\|\hat{m}_0 - m_{0,n}\|_{2,P_n} = O_P(a_{0,n})$, and, uniformly in $t \in [0, 1]$,*

$$\mathbb{P}_n\{\psi_t^{\text{PN}}(O; \hat{\eta}) - \psi_t^{\text{PN}}(O; \eta)\} = O_P\left(\frac{a_{0,n}}{\sqrt{n}} + a_{e,n}\sqrt{\frac{E[m_{0,n}(X)]}{n}} + a_{e,n}a_{0,n}\right).$$

Conditionally on training, the score-difference mean is $E[\delta_e(X)\delta_0(X)/\{1 - \hat{e}(X)\}]$, and its variance is at most $C_\varepsilon\{a_{0,n}^2 + a_{e,n}^2 E[m_{0,n}(X)]\}$ on the learning event.

Proof. Fix a held-out fold and condition on its training sample. Suppress row and fold indices, put $m = m_{0,n}$, $\hat{m} = \hat{m}_{0,-k,n}$, $e = e_n$, $\hat{e} = \hat{e}_{-k,n}$, $\delta_0 = \hat{m} - m$, and $\delta_e = \hat{e} - e$. The two tail lemmas in the dependency list imply, outside an event of probability $O(n^{-M})$ for every fixed M ,

$$\|\delta_e\|_\infty \vee \|\delta_e\|_{2,P_n} \leq C_M a_{e,n}, \quad \|\delta_0\|_{2,P_n} \leq C_M a_{0,n}. \quad (1)$$

The score difference does not depend on t , because the term $-tAY$ cancels. Direct subtraction in `def:scores` gives

$$\Delta = -A\delta_0 + \frac{\hat{e}}{1 - \hat{e}}(1 - A)\delta_0 - \left\{ \frac{\hat{e}}{1 - \hat{e}} - \frac{e}{1 - e} \right\} (1 - A)(Y - m). \quad (2)$$

Conditional expectation of the last term is zero, and elementary algebra in the first two terms yields

$$E(\Delta \mid \text{training}) = E\left[\frac{\delta_e(X)\delta_0(X)}{1 - \widehat{e}(X)}\right]. \quad (3)$$

The fitted propensity lies in $[\varepsilon/2, 1 - \varepsilon/2]$, so Cauchy–Schwarz and (1) bound (3) by $C_\varepsilon a_{e,n} a_{0,n}$. Also, the odds map has a bounded derivative on the common overlap interval. Squaring (2), conditioning on X, A , and using $\text{Var}(Y \mid A = 0, X) \leq m(X)$ gives

$$\text{Var}(\Delta \mid \text{training}) \leq E(\Delta^2 \mid \text{training}) \leq C_\varepsilon \{a_{0,n}^2 + a_{e,n}^2 E[m_{0,n}(X)]\}. \quad (4)$$

For each held-out fold, conditional Chebyshev applied to its average, followed by (3)–(4), gives

$$\mathbb{P}_{n,k}\Delta = O_P\left(\frac{a_{0,n}}{\sqrt{n}} + a_{e,n}\sqrt{\frac{E[m_{0,n}(X)]}{n}} + a_{e,n}a_{0,n}\right). \quad (5)$$

The number of balanced folds is fixed, so averaging (5) over folds and taking a union bound proves the pooled assertion. Because Δ is independent of t , the assertion is uniform on $[0, 1]$. The learning rates in the first sentence are exactly (1), completing every part of the lemma. $\square$

Lemma 20 (lem:ps-score-learning-remainder). *On the PS upper class, the PS learner satisfies $\|\widehat{e} - e_n\|_\infty \vee \|\widehat{e} - e_n\|_{2,P_n} = O_P(a_{e,n})$, $\|\widehat{m}_1 - m_{1,n}\|_{2,P_n} = O_P(a_{1,n})$, and, uniformly in $t \in [0, 1]$,*

$$\mathbb{P}_n\{\psi_t^{\text{PS}}(O; \widehat{\eta}^{\text{PS}}) - \psi_t^{\text{PS}}(O; \eta^{\text{PS}})\} = O_P\left(\frac{a_{1,n}}{\sqrt{n}} + a_{e,n}\sqrt{\frac{E[1 - m_{1,n}(X)]}{n}} + a_{e,n}a_{1,n}\right).$$

The corresponding conditional second moment is at most $C_\varepsilon \{a_{1,n}^2 + a_{e,n}^2 E[1 - m_{1,n}(X)]\}$ on the learning event.

Proof. Apply the PN algebra to the transformed variables $A^\dagger = 1 - A$, $Y^\dagger = 1 - Y$, propensity $\pi = 1 - e_n$, outcome regression $\mu = 1 - m_{1,n}$, fitted propensity $\widehat{\pi} = 1 - \widehat{e}$, and fitted regression $\widehat{\mu} = 1 - \widehat{m}_1$. This is exactly the PS score by `def:scores`. The two learning-tail lemmas give

$$\|\widehat{\pi} - \pi\|_\infty \vee \|\widehat{\pi} - \pi\|_{2,P_n} = O_P(a_{e,n}), \quad \|\widehat{\mu} - \mu\|_{2,P_n} = O_P(a_{1,n}). \quad (1)$$

Writing $\delta_\pi = \widehat{\pi} - \pi$ and $\delta_\mu = \widehat{\mu} - \mu$, direct subtraction gives a difference independent of t , with conditional mean

$$E\left[\frac{\delta_\pi(X)\delta_\mu(X)}{1 - \widehat{\pi}(X)}\right], \quad (2)$$

and conditional second moment at most

$$C_\varepsilon \{\|\delta_\mu\|_{2,P_n}^2 + \|\delta_\pi\|_\infty^2 E[\mu(X)]\} \leq C_\varepsilon \{a_{1,n}^2 + a_{e,n}^2 E[1 - m_{1,n}(X)]\}. \quad (3)$$

Here (3) uses $\text{Var}(Y^\dagger \mid A^\dagger = 0, X) \leq \mu(X)$ and the common clipped-overlap bound. Conditional Chebyshev for each held-out fold, Cauchy–Schwarz in (2), and a union bound over fixed K yield

$$\mathbb{P}_n\{\psi_t^{\text{PS}}(O; \widehat{\eta}^{\text{PS}}) - \psi_t^{\text{PS}}(O; \eta^{\text{PS}})\} = O_P\left(\frac{a_{1,n}}{\sqrt{n}} + a_{e,n}\sqrt{\frac{E[1 - m_{1,n}(X)]}{n}} + a_{e,n}a_{1,n}\right).$$

The cancellation of the affine t -term makes this uniform over $t \in [0, 1]$, and (1)–(3) prove all remaining assertions. $\square$

Proposition 3 (prop:pn-causal-localization). *On $\mathcal{B}_n^{\text{PN}}$, $m_{0,n}(X) \leq m_{1,n}(X)$ $P_{X,n}$ -almost surely and $E[m_{0,n}(X)] \leq E[m_{1,n}(X)] \leq q_{\text{PN}}(P_n)/\varepsilon$. Consequently the PN fitted-score remainder is $O_P(a_{0,n}/\sqrt{n} + a_{e,n}\sqrt{q_{\text{PN}}(P_n)/n} + a_{e,n}a_{0,n})$, and on every $(r_n) \in \mathcal{R}^{\text{PN}}$ it is $o_P(\sqrt{r_n/n})$.*

Proof. By **lem:generic**, $m_{0,n}(X) \leq m_{1,n}(X)$ almost surely. Since $e_n(X) \geq \varepsilon$,

$$q_{\text{PN}}(P_n) = E[e_n(X)m_{1,n}(X)] \geq \varepsilon E[m_{1,n}(X)],$$

which proves $E[m_{0,n}(X)] \leq E[m_{1,n}(X)] \leq q_{\text{PN}}(P_n)/\varepsilon$. Substitution of this last bound into **lem:pn-score-learning-remainder** gives

$$O_P\left(\frac{a_{0,n}}{\sqrt{n}} + a_{e,n}\sqrt{\frac{q_{\text{PN}}(P_n)}{n}} + a_{e,n}a_{0,n}\right).$$

On a PN Gaussian shell, $q_{\text{PN}}(P_n) = r_n$. Dividing the three terms by $\sqrt{r_n/n}$ gives respectively $a_{0,n}/\sqrt{r_n}$, $a_{e,n}$, and $\{na_{e,n}^2a_{0,n}^2/r_n\}^{1/2}$. The first tends to zero by **lem:upper-schedule-consequences**, the second because $a_{e,n} \rightarrow 0$, and the third by the defining membership $(r_n) \in \mathcal{R}^{\text{PN}}$. This proves the asserted uniform little-oh remainder. $\square$

Proposition 4 (**prop:ps-causal-localization**). *On $\mathcal{B}_n^{\text{PS}}$, $1 - m_{1,n}(X) \leq 1 - m_{0,n}(X)$ $P_{X,n}$ -almost surely and $E[1 - m_{1,n}(X)] \leq E[1 - m_{0,n}(X)] \leq q_{\text{PS}}(P_n)/\varepsilon$. Consequently the PS fitted-score remainder is $O_P(a_{1,n}/\sqrt{n} + a_{e,n}\sqrt{q_{\text{PS}}(P_n)/n} + a_{e,n}a_{1,n})$, and on every $(r_n) \in \mathcal{R}^{\text{PS}}$ it is $o_P(\sqrt{r_n/n})$.*

Proof. By **lem:generic**, $m_{0,n} \leq m_{1,n}$, hence $1 - m_{1,n} \leq 1 - m_{0,n}$ almost surely. Overlap gives

$$q_{\text{PS}}(P_n) = E[(1 - e_n(X))\{1 - m_{0,n}(X)\}] \geq \varepsilon E[1 - m_{0,n}(X)].$$

Therefore $E[1 - m_{1,n}(X)] \leq E[1 - m_{0,n}(X)] \leq q_{\text{PS}}(P_n)/\varepsilon$. Substitution in **lem:ps-score-learning-remainder** gives the displayed PS remainder. On the PS shell, divide its three terms by $\sqrt{r_n/n}$. The ratios are $a_{1,n}/\sqrt{r_n} \rightarrow 0$, $a_{e,n} \rightarrow 0$, and $\{na_{e,n}^2a_{1,n}^2/r_n\}^{1/2} \rightarrow 0$, by **lem:upper-schedule-consequences** and the defining membership $(r_n) \in \mathcal{R}^{\text{PS}}$. This proves the asserted little-oh statement uniformly on the shell. $\square$

Theorem 4 (**thm:uncapped-ultra-rare-boundary-score-robustness**). *Fix $\alpha \in (0, 1/2)$, $d \in \{\text{PN}, \text{PS}\}$, and $c_G > 0$. For every schedule $0 < r_n \leq 1$ satisfying $nr_n \rightarrow \infty$ and $nr_n a_{e,n}^2 \rightarrow 0$, suppose $\mathcal{S}_{G,n}^{\partial,d}(r_n; c_G) \neq \emptyset$ eventually. The propensity-only inversion $\bar{\mathcal{C}}_{n,\alpha}^d$ in **def:boundary-score-inversion** then has uniform asymptotic coverage at least $1 - \alpha$ and*

$$\sup_{P_n \in \mathcal{S}_{G,n}^{\partial,d}(r_n; c_G)} E_{P_n^{\otimes n}}[\text{diam}(\bar{\mathcal{C}}_{n,\alpha}^d)] = O((nr_n)^{-1/2}).$$

No outcome-regression smoothness or least-favorable cap is imposed.

Proof. Use one transformed notation for both directions. For PN put $A^* = A$, $Y^* = Y$, $\pi = e_n$, $\mu_0 = m_{0,n}$, and $\mu_1 = m_{1,n}$. For PS put $A^* = 1 - A$, $Y^* = 1 - Y$, $\pi = 1 - e_n$, $\mu_0 = 1 - m_{1,n}$, and $\mu_1 = 1 - m_{0,n}$. Then in either case $0 \leq \mu_0 \leq \mu_1 \leq 1$, $\varepsilon \leq \pi \leq 1 - \varepsilon$,

$$U = A^*Y^*, \quad J = (1 - A^*)Y^*, \quad EU = r_n, \quad \theta_n = 1 - \frac{E[\pi(X)\mu_0(X)]}{r_n}. \quad (1)$$

The last identity is **def:functionals**. The oracle affine score

$$\phi_t = (1 - t)U - \frac{\pi(X)}{1 - \pi(X)}J \quad (2)$$

is centered at $t = \theta_n$. This is also the score identity in **lem:identification**. Since $r_n = E[\pi\mu_1] \geq \varepsilon E\mu_1$,

$$E\mu_0 \leq E\mu_1 \leq r_n/\varepsilon, \quad E(U + J) \leq C_\varepsilon r_n. \quad (3)$$

Consequently $E\phi_{\theta_n}^2 \leq C_\varepsilon r_n$. Let $\psi_{\theta_n}^d$ be the frozen orthogonal score and set $S = \phi_{\theta_n} - \psi_{\theta_n}^d = \mu_0(X)\{A^* - \pi(X)(1 - \pi(X))^{-1}(1 - A^*)\}$. Then $ES^2 = E[\pi(1 - \pi)^{-1}\mu_0^2] \leq \varepsilon^{-1}E\phi_{\theta_n}^2$, because the control-arm contribution to $E\phi_{\theta_n}^2$ is $E[\pi^2(1 - \pi)^{-1}\mu_0]$. Hence $V_d(P_n) = E(\phi_{\theta_n} - S)^2 \leq 2(1 + \varepsilon^{-1})E\phi_{\theta_n}^2$, and the raw-shell condition implies $E\phi_{\theta_n}^2 \geq c_G r_n / \{2(1 + \varepsilon^{-1})\}$. Define $V_d^*(P_n) = E\phi_{\theta_n}^2$; thus $c_* r_n \leq V_d^*(P_n) \leq \bar{C}_\varepsilon r_n$, where $c_* = c_G / \{2(1 + \varepsilon^{-1})\} > 0$.

Let $h(u) = u/(1-u)$, and let $\hat{\pi}$ denote the cross-fitted propensity in the selected direction. The fitted-minus-oracle difference at the truth is

$$R_i = -\{h(\hat{\pi}_{-k(i)}(X_i)) - h(\pi(X_i))\}J_i. \quad (4)$$

On the propensity-learning event, $\|\hat{\pi} - \pi\|_\infty \leq C_M a_{e,n}$. Since fitted and true propensities share a fixed overlap interval, conditioning on a training sample gives

$$|E(R_i \mid \text{training})| \leq C a_{e,n} E\mu_0 \leq C a_{e,n} r_n, \quad E(R_i^2 \mid \text{training}) \leq C a_{e,n}^2 E J \leq C a_{e,n}^2 r_n. \quad (5)$$

A fixed-fold conditional Chebyshev argument and the polynomial learning tails imply

$$\bar{g}_{d,n}(\theta_n) = \mathbb{P}_n \phi_{\theta_n} + O_P(a_{e,n} r_n) + O_P\left(a_{e,n} \sqrt{\frac{r_n}{n}}\right) = \mathbb{P}_n \phi_{\theta_n} + o_P\left(\sqrt{\frac{r_n}{n}}\right), \quad (6)$$

because $nr_n a_{e,n}^2 \rightarrow 0$. The same calculation gives $\mathbb{P}_n R^2 = o_P(r_n)$. Bounded exponential-moment estimates applied to the oracle first and second moments, using (3) and $nr_n \rightarrow \infty$, give

$$\mathbb{P}_n \phi_{\theta_n}^2 = V_d^*(P_n) + o_P(r_n), \quad \mathbb{P}_n \phi_{\theta_n} = O_P(\sqrt{r_n/n}). \quad (7)$$

The empirical-variance comparison inequality and (6)–(7) therefore yield

$$\frac{\bar{\sigma}_{d,n}^2(\theta_n)}{V_d^*(P_n)} \rightarrow_P 1 \quad (8)$$

uniformly over the raw shell.

The oracle variables in (2) are iid, centered, bounded by a constant depending only on ε , and have variance between $c_* r_n$ and $C_\varepsilon r_n$. Since $nr_n \rightarrow \infty$, `lem:bounded-iid-normal-approximation`, (6), and (8) imply the uniform studentized limit

$$\frac{\sqrt{n} \bar{g}_{d,n}(\theta_n)}{\bar{\sigma}_{d,n}(\theta_n)} \rightsquigarrow N(0, 1). \quad (9)$$

Thus $P_n^{\otimes n} \{\theta_n \in \bar{\mathcal{A}}_{n,\alpha}^d\} \rightarrow 1 - \alpha$ uniformly. The singleton fallback preserves this event, proving the coverage assertion.

It remains to bound diameter in expectation. Put $\hat{r}_n = \mathbb{P}_n U$ and $W = U + J$. Affinity in t gives

$$\bar{g}_{d,n}(t) - \bar{g}_{d,n}(s) = -(t - s)\hat{r}_n. \quad (10)$$

Clipped overlap also gives $\sup_t \bar{\sigma}_{d,n}^2(t) \leq C_\varepsilon \mathbb{P}_n W$. The bounded exponential-moment inequality, $EU = r_n$, and (3) imply

$$P\{\hat{r}_n < r_n/2\} + P\{\mathbb{P}_n W > C_\varepsilon r_n\} \leq C \exp(-c n r_n). \quad (11)$$

On the complement of the events in (11), any accepted s, t satisfy, by (10),

$$|t - s| \frac{r_n}{2} \leq |\bar{g}_{d,n}(t)| + |\bar{g}_{d,n}(s)| \leq \frac{2z_{1-\alpha/2}}{\sqrt{n}} \sup_u \bar{\sigma}_{d,n}(u) \leq C \sqrt{\frac{r_n}{n}}. \quad (12)$$

Hence the preliminary diameter is at most $C(nr_n)^{-1/2}$ on the good event. It is always at most one, and the fallback has diameter zero when the preliminary set is empty. Since $e^{-c n r_n} = O((nr_n)^{-1/2})$, taking expectations in (11)–(12) proves the asserted uniform expected-diameter rate. No bound above used an outcome-regression Hölder condition or the cap $\bar{d}$. $\square$

Justification. The propensity-only affine score supplies an outcome-smoothness-free internal robustness result when $nr_n a_{e,n}^2 \rightarrow 0$. *Gap.* It is not positioned as a field-level necessity theorem and supplies no converse of its own. *Consumer.* It provides an alternative score on verified raw boundary shells.

Lemma 21 (`lem:rare-score-empirical-envelope`). *Fix a direction d , an admissible schedule (r_n) , and its Gaussian shell. For every fixed $M > 0$, there are constants $C_M, c_M > 0$, depending only on the frozen primitive tuple and M , such that, uniformly over $P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)$,*

$$P_n^{\otimes n} \left\{ \sup_{t \in [0,1]} \mathbb{P}_n [\psi_{t,n}^d(O; \hat{\eta}_{d,-k,n})^2] > C_M r_n \right\} \leq C_M n^{-M} + C_M e^{-c_M n r_n}$$

for all sufficiently large n , where $\mathbb{P}_n$ is the pooled held-out average.

Proof. If $d = \text{PN}$, the assertion is exactly **lem:pn-rare-score-empirical-envelope**; the learner and shell in the display are the PN learner and PN shell. If $d = \text{PS}$, it is exactly **lem:ps-rare-score-empirical-envelope**. The schedule and shell definitions select precisely one of these two cases, so the corresponding constants depend only on the fixed primitive tuple and M . This proves the unified statement. $\square$

Lemma 22 (lem:pn-variance-consistency). *Uniformly on $\mathcal{M}_{G,n}^{\text{PN}}(r_n; c_G, C_G)$, at $t_n = \Theta_{\text{PN}}(P_n)$, $\mathbb{P}_n \Delta_{\text{PN}}^2 = O_P(a_{0,n}^2 + r_n a_{e,n}^2) = o_P(r_n)$, $\hat{\sigma}_{\text{PN},n}^2(t_n)/V_{\text{PN}}(P_n) \rightarrow_P 1$, and $B_\varepsilon/\sqrt{nr_n} \rightarrow 0$.*

Proof. Put $t_n = \Theta_{\text{PN}}(P_n)$, let $\Psi_i = \psi_{t_n,n}^{\text{PN}}(O_{i,n}; \eta_{i,n}^{\text{PN}})$, and let Δ_i be its fitted-minus-oracle held-out difference. The conditional second-moment calculation in **lem:pn-score-learning-remainder**, together with **prop:pn-causal-localization**, gives on the uniform learning event

$$E(\Delta_i^2 \mid \text{training}) \leq C\{a_{0,n}^2 + r_n a_{e,n}^2\}. \quad (1)$$

Conditional Markov for each fold and a fixed-fold union bound therefore yield

$$\mathbb{P}_n \Delta_{\text{PN}}^2 = O_P(a_{0,n}^2 + r_n a_{e,n}^2) = o_P(r_n), \quad (2)$$

because $a_{0,n}^2/r_n \rightarrow 0$ by **lem:upper-schedule-consequences** and $a_{e,n} \rightarrow 0$.

By **lem:identification**, $E\Psi_i = 0$. The structural upper bound and shell lower bound give

$$c_G r_n \leq V_{\text{PN}}(P_n) = E\Psi_i^2 \leq \varepsilon^{-1} r_n. \quad (3)$$

The oracle score is bounded by B_ε . The elementary bounded exponential-moment inequality applied to $\Psi_i^2 - E\Psi_i^2$, whose variance is at most $B_\varepsilon^2 E\Psi_i^2$, gives

$$\mathbb{P}_n \Psi^2 - E\Psi^2 = O_P\left(\sqrt{\frac{r_n}{n}} + \frac{1}{n}\right) = o_P(r_n), \quad \mathbb{P}_n \Psi = O_P\left(\sqrt{\frac{r_n}{n}}\right). \quad (4)$$

The last equalities use $nr_n \rightarrow \infty$. Hence the empirical variance of the oracle scores equals $V_{\text{PN}}(P_n) + o_P(r_n)$. For any vectors x, y , the difference of their empirical centered second moments is bounded by

$$\mathbb{P}_n y^2 + 2\{\mathbb{P}_n(x - \mathbb{P}_n x)^2\}^{1/2}\{\mathbb{P}_n(y - \mathbb{P}_n y)^2\}^{1/2}. \quad (5)$$

Apply (5) with $x = \Psi$, $y = \Delta$, and use (2)–(4). The fitted empirical variance is $V_{\text{PN}}(P_n) + o_P(r_n)$; division by the lower bound in (3) proves the ratio consistency. Finally $B_\varepsilon/\sqrt{nr_n} \rightarrow 0$ follows from $nr_n \rightarrow \infty$. All bounds are uniform because the learning events and constants in the cited nodes are uniform over the shell. $\square$

Lemma 23 (lem:ps-variance-consistency). *Uniformly on $\mathcal{M}_{G,n}^{\text{PS}}(r_n; c_G, C_G)$, at $t_n = \Theta_{\text{PS}}(P_n)$, $\mathbb{P}_n \Delta_{\text{PS}}^2 = O_P(a_{1,n}^2 + r_n a_{e,n}^2) = o_P(r_n)$, $\hat{\sigma}_{\text{PS},n}^2(t_n)/V_{\text{PS}}(P_n) \rightarrow_P 1$, and $B_\varepsilon/\sqrt{nr_n} \rightarrow 0$.*

Proof. Transform to $A^\dagger = 1 - A$, $Y^\dagger = 1 - Y$, so that the PS score is the PN score with fitted-minus-oracle difference Δ_{PS} . By **lem:ps-score-learning-remainder** and **prop:ps-causal-localization**,

$$\mathbb{P}_n \Delta_{\text{PS}}^2 = O_P(a_{1,n}^2 + r_n a_{e,n}^2) = o_P(r_n). \quad (1)$$

The last equality uses $a_{1,n}^2/r_n \rightarrow 0$ and $a_{e,n} \rightarrow 0$. Identification centers the oracle score, while the PS shell lower bound and **lem:ps-structural-score-variance-upper-bound** give

$$c_G r_n \leq V_{\text{PS}}(P_n) \leq \varepsilon^{-1} r_n. \quad (2)$$

Because the oracle score is bounded, the bounded exponential-moment calculation for its first two empirical moments gives

$$\mathbb{P}_n \Psi^2 = V_{\text{PS}}(P_n) + o_P(r_n), \quad \mathbb{P}_n \Psi = O_P(\sqrt{r_n/n}), \quad (3)$$

uniformly, using $nr_n \rightarrow \infty$. The empirical-variance comparison inequality

$$|\widehat{\text{Var}}(\Psi + \Delta) - \widehat{\text{Var}}(\Psi)| \leq \mathbb{P}_n \Delta^2 + 2\{\widehat{\text{Var}}(\Psi)\mathbb{P}_n \Delta^2\}^{1/2}$$

and (1)–(3) show that the difference is $o_P(r_n)$. Division by (2) proves $\hat{\sigma}_{\text{PS},n}^2(t_n)/V_{\text{PS}}(P_n) \rightarrow_P 1$. The schedule consequence $nr_n \rightarrow \infty$ also gives $B_\varepsilon/\sqrt{nr_n} \rightarrow 0$. $\square$

Lemma 24 (lem:pn-rare-score-empirical-envelope). *For every fixed $M > 0$, uniformly over the PN Gaussian shell,*

$$P_n^{\otimes n} \left\{ \sup_{t \in [0,1]} \mathbb{P}_n[\psi_{t,n}^{\text{PN}}(O; \hat{\eta}_{-k,n}^{\text{PN}})^2] > C_M r_n \right\} \leq C_M n^{-M} + C_M e^{-c_M n r_n}$$

for constants $C_M, c_M > 0$.

Proof. On the simultaneous learning event from `lem:uniform-propensity-spline-learning-tails` and `lem:control-spline-learning-tails`, write $\hat{m}_0 = m_{0,n} + \delta_0$. Casewise evaluation of `def:scores`, using the clipped odds bound, gives for every $t \in [0, 1]$

$$\{\psi_{t,n}^{\text{PN}}(O; \hat{\eta})\}^2 \leq C_\varepsilon \{Y + m_{0,n}(X) + \delta_0(X)^2\}. \quad (1)$$

By `prop:pn-causal-localization`, $E[m_{0,n}(X)] \leq E[m_{1,n}(X)] \leq r_n/\varepsilon$, and overlap also gives $E[Y] \leq C_\varepsilon r_n$. On the learning event, $E\delta_0(X)^2 \leq C_M a_{0,n}^2 = o(r_n)$. For each held-out fold, the three variables on the right of (1) are bounded. The elementary bounded exponential-moment inequality therefore gives, conditionally on training,

$$P\{\mathbb{P}_{n,k}[Y + m_{0,n}(X) + \delta_0(X)^2] > C_M r_n \mid \text{training}\} \leq C_M e^{-c_M n r_n}. \quad (2)$$

For the last summand, use $E\delta_0^2 = o(r_n)$; its boundedness makes the exponent at threshold r_n at least a constant multiple of $n r_n$. Taking a union bound over fixed folds and adding the $C_M n^{-M}$ learning-event failure probability proves the claimed inequality. The deterministic domination (1) already takes the supremum over t . $\square$

Lemma 25 (lem:ps-rare-score-empirical-envelope). *For every fixed $M > 0$, uniformly over the PS Gaussian shell,*

$$P_n^{\otimes n} \left\{ \sup_{t \in [0,1]} \mathbb{P}_n[\psi_{t,n}^{\text{PS}}(O; \hat{\eta}_{-k,n}^{\text{PS}})^2] > C_M r_n \right\} \leq C_M n^{-M} + C_M e^{-c_M n r_n}$$

for constants $C_M, c_M > 0$.

Proof. Use $A^\dagger = 1 - A$, $Y^\dagger = 1 - Y$, $\mu = 1 - m_{1,n}$, and $\hat{\mu} = 1 - \hat{m}_{1,n} = \mu + \delta_1^\dagger$. The clipped PS score obeys, uniformly over $t \in [0, 1]$,

$$\{\psi_{t,n}^{\text{PS}}(O; \hat{\eta}^{\text{PS}})\}^2 \leq C_\varepsilon \{Y^\dagger + \mu(X) + \delta_1^\dagger(X)^2\}. \quad (1)$$

The PS localization inequalities give $E\mu(X) \leq r_n/\varepsilon$ and $E[Y^\dagger] \leq C_\varepsilon r_n$. On the treated-learning event, $E(\delta_1^\dagger)^2 \leq C_M a_{1,n}^2 = o(r_n)$. Conditional bounded exponential-moment bounds at threshold $C_M r_n$ consequently have failure probability at most $C_M e^{-c_M n r_n}$ on each held-out fold. Add the $C_M n^{-M}$ probability of leaving either learning event and take a union bound over fixed K . The result follows from (1). $\square$

Theorem 5 (thm:pn-gaussian-frontier). *Fix $\alpha \in (0, 1/2)$, $c_G > 0$, $C_G = \varepsilon^{-1}$, and $(r_n) \in \mathcal{R}^{\text{PN}}$. If the PN Gaussian shells are eventually nonempty, then uniformly over them the PN score has its oracle expansion with remainder $o_P(\sqrt{r_n/n})$, its studentized value at $\Theta_{\text{PN}}(P_n)$ converges to $N(0, 1)$, its inversion has coverage at least $1 - \alpha$, and its expected diameter is $O((n r_n)^{-1/2})$.*

Proof. Let $t_n = \Theta_{\text{PN}}(P_n)$. The remainder conclusion of `prop:pn-causal-localization` gives the uniform oracle expansion

$$\hat{g}_{\text{PN},n}(t_n) = \mathbb{P}_n \psi_{t_n,n}^{\text{PN}}(O; \eta_n^{\text{PN}}) + o_P(\sqrt{r_n/n}). \quad (1)$$

By identification the oracle summands are centered. Their variance lies in $[c_G r_n, \varepsilon^{-1} r_n]$, by the shell definition and `lem:pn-structural-score-variance-upper-bound`, and they are bounded by B_ε . Since $n r_n \rightarrow \infty$, the hypotheses of `lem:bounded-score-studentization` are satisfied by (1) and `lem:pn-variance-consistency`. Hence

$$\frac{\sqrt{n} \hat{g}_{\text{PN},n}(t_n)}{\hat{\sigma}_{\text{PN},n}(t_n)} \rightsquigarrow N(0, 1) \quad (2)$$

uniformly. The event $|(2)| \leq z_{1-\alpha/2}$ is exactly $t_n \in \mathcal{A}_{n,\alpha}^{\text{PN}}$, so the preliminary set has limiting coverage $1 - \alpha$, and the fallback preserves it.

For diameter, put $\hat{r}_n = \mathbb{P}_n(AY)$. Affinity of the score gives $\hat{g}(t) - \hat{g}(s) = -(t - s)\hat{r}_n$. The factual indicator is Bernoulli with mean r_n , so a bounded exponential-moment bound gives $P(\hat{r}_n < r_n/2) \leq e^{-c n r_n}$. By `lem:pn-rare-score-empirical-envelope`, outside probability $C_M n^{-M} + C_M e^{-c_M n r_n}$,

$$\sup_{u \in [0,1]} \hat{\sigma}_{\text{PN},n}^2(u) \leq \sup_u \mathbb{P}_n \psi_{u,n}^{\text{PN}}(\hat{\eta})^2 \leq C_M r_n. \quad (3)$$

On the intersection of these events, any $s, t \in \mathcal{A}_{n,\alpha}^{\text{PN}}$ obey

$$|t - s| \frac{r_n}{2} \leq |\hat{g}(t)| + |\hat{g}(s)| \leq \frac{2z_{1-\alpha/2}}{\sqrt{n}} \sup_u \hat{\sigma}(u) \leq C \sqrt{r_n/n}. \quad (4)$$

Thus the diameter is at most $C(nr_n)^{-1/2}$ there. It is always at most one, and the fallback cannot enlarge it. Taking $M \geq 1$, both n^{-M} and $e^{-c n r_n}$ are $O((nr_n)^{-1/2})$; taking expectations in (4) proves the claimed rate. $\square$

Theorem 6 (`thm:ps-gaussian-frontier`). *Fix $\alpha \in (0, 1/2)$, $c_G > 0$, $C_G = \varepsilon^{-1}$, and $(r_n) \in \mathcal{R}^{\text{PS}}$. If the PS Gaussian shells are eventually nonempty, then uniformly over them the PS score has its oracle expansion with remainder $o_P(\sqrt{r_n/n})$, its studentized value at $\Theta_{\text{PS}}(P_n)$ converges to $N(0, 1)$, its inversion has coverage at least $1 - \alpha$, and its expected diameter is $O((nr_n)^{-1/2})$.*

Proof. Let $t_n = \Theta_{\text{PS}}(P_n)$. The PS localization proposition gives

$$\hat{g}_{\text{PS},n}(t_n) = \mathbb{P}_n \psi_{t_n,n}^{\text{PS}}(O; \eta_n^{\text{PS}}) + o_P(\sqrt{r_n/n}) \quad (1)$$

uniformly. Identification centers the oracle score; its variance is between $c_G r_n$ and $\varepsilon^{-1} r_n$, and its absolute value is bounded by B_ε . Together with $nr_n \rightarrow \infty$, `lem:ps-variance-consistency` and `lem:bounded-score-studentization` therefore prove uniform studentized convergence to $N(0, 1)$. Inversion at the true value gives limiting coverage $1 - \alpha$, and the singleton fallback preserves coverage.

Put $\hat{r}_n = \mathbb{P}_n[(1 - A)(1 - Y)]$. The PS score is affine with slope $-\hat{r}_n$. The Bernoulli lower-tail bound gives $P(\hat{r}_n < r_n/2) \leq e^{-c n r_n}$, and `lem:ps-rare-score-empirical-envelope` gives $\sup_t \hat{\sigma}_{\text{PS},n}^2(t) \leq C r_n$ outside probability $C_M n^{-M} + C_M e^{-c_M n r_n}$. Thus two accepted points s, t satisfy

$$|s - t| \frac{r_n}{2} \leq \frac{2z_{1-\alpha/2}}{\sqrt{n}} \sup_u \hat{\sigma}_{\text{PS},n}(u) \leq C \sqrt{r_n/n}.$$

The diameter is bounded by one on the exceptional event and is not enlarged by the fallback. Taking expectations, with $nr_n \rightarrow \infty$, proves $O((nr_n)^{-1/2})$. $\square$

Theorem 7 (`thm:ultra-rare-boundary-score-robustness`). *Fix $\alpha \in (0, 1/2)$, $d \in \{\text{PN}, \text{PS}\}$, any fixed $c_G > 0$, and every deterministic schedule $0 < r_n \leq \bar{d}$ satisfying*

$$nr_n \rightarrow \infty, \quad nr_n a_{e,n}^2 \rightarrow 0.$$

Define

d	$U_d^\star(O)$	$J_d^\star(O)$
PN	AY	$(1 - A)Y$
PS	$(1 - A)(1 - Y)$	$A(1 - Y)$

and put $\pi_{\text{PN}} = e$, $\pi_{\text{PS}} = 1 - e$, $\hat{\pi}_{\text{PN},-k} = \hat{e}_{-k}$, and $\hat{\pi}_{\text{PS},-k} = 1 - \hat{e}_{-k}$. Let

$$\bar{\phi}_{t,-k}^d(O) = (1 - t)U_d^\star(O) - \frac{\hat{\pi}_{d,-k}(X)}{1 - \hat{\pi}_{d,-k}(X)} J_d^\star(O),$$

$$\bar{g}_{d,n}(t) = \frac{1}{n} \sum_{i=1}^n \bar{\phi}_{t,-k(i)}^d(O_{i,n}), \quad \bar{\sigma}_{d,n}^2(t) = \frac{1}{n} \sum_{i=1}^n \{\bar{\phi}_{t,-k(i)}^d(O_{i,n}) - \bar{g}_{d,n}(t)\}^2.$$

Define

$$\bar{\mathcal{A}}_{n,\alpha}^d = \{t \in [0, 1] : n\bar{g}_{d,n}(t)^2 \leq z_{1-\alpha/2}^2 \bar{\sigma}_{d,n}^2(t)\}$$

, and report $\bar{\mathcal{C}}_{n,\alpha}^d = \bar{\mathcal{A}}_{n,\alpha}^d$ when this set is nonempty and $\bar{\mathcal{C}}_{n,\alpha}^d = \{0\}$ otherwise. Suppose $\mathcal{S}_{G,n}^{\partial,d}(r_n; c_G) \neq \emptyset$ for all sufficiently large n . Uniformly over this propensity-only boundary shell,

$$\liminf_{n \rightarrow \infty} \inf_{P_n \in \mathcal{S}_{G,n}^{\partial,d}(r_n; c_G)} P_n^{\otimes n} \{\Theta_d(P_n) \in \bar{\mathcal{C}}_{n,\alpha}^d\} \geq 1 - \alpha$$

and

$$\sup_{P_n \in \mathcal{S}_{G,n}^{\partial,d}(r_n; c_G)} E_{P_n^{\otimes n}} [\text{diam}(\bar{\mathcal{C}}_{n,\alpha}^d)] = O((nr_n)^{-1/2}).$$

Writing $b = b(d)$, choose

$$\max\{1 - 2s_e, 2s_b, 2s_e + 2s_b - 1\} < \gamma < 1, \quad r_n = \bar{d} \wedge n^{-\gamma}.$$

Then

$$nr_n \rightarrow \infty, \quad nr_n a_{e,n}^2 \rightarrow 0, \quad \frac{r_n}{a_{b,n}^2} \rightarrow 0, \quad \frac{r_n}{na_{e,n}^2 a_{b,n}^2} \rightarrow 0.$$

For LF-compatible primitives and $0 < c_G \leq 3/16$, **lem:boundary-shell-nonempty-witness** makes every shell in this display nonempty. Thus an alternative propensity-only first-order inversion remains honest and shrinking on explicit schedules outside the sufficient fixed two-nuisance spline regime. This is an internal robustness construction: it refutes no claim in the literature and asserts no negative literature novelty.

Proof. For either direction make the transformation $(A, Y, e, m_0, m_1) \mapsto (A^*, Y^*, \pi, \mu_0, \mu_1)$ used in the preceding boundary argument: it is the identity for PN and $(1 - A, 1 - Y, 1 - e, 1 - m_1, 1 - m_0)$ for PS. Monotonicity and overlap give $0 \leq \mu_0 \leq \mu_1 \leq 1$ and $\varepsilon \leq \pi \leq 1 - \varepsilon$. With $U = A^* Y^*$, $J = (1 - A^*) Y^*$, and $q_d(P_n) = r_n$,

$$\theta_n = \Theta_d(P_n) = 1 - \frac{E[\pi \mu_0]}{r_n}, \quad E[(1 - \theta_n)U - \pi(1 - \pi)^{-1}J] = 0. \quad (1)$$

Denote the oracle affine score by $\phi_t = (1 - t)U - \pi(1 - \pi)^{-1}J$. Moreover $E\mu_0 \leq E\mu_1 \leq r_n/\varepsilon$, so the affine oracle-score variance is at most $C_\varepsilon r_n$. If $S = \mu_0(X)\{A^* - \pi(X)(1 - \pi(X))^{-1}(1 - A^*)\}$, then the frozen orthogonal score equals the affine score minus S , while $ES^2 \leq \varepsilon^{-1}E\phi_{\theta_n}^2$. Thus $V_d(P_n) \leq 2(1 + \varepsilon^{-1})E\phi_{\theta_n}^2$, and the raw-shell lower bound yields affine-score variance at least $c_G r_n / \{2(1 + \varepsilon^{-1})\}$. Write $V_d^*(P_n) = E\phi_{\theta_n}^2$ for this affine-score variance.

For the cross-fitted score, the only fitted-minus-oracle term is $-\{h(\hat{\pi}) - h(\pi)\}J$, where $h(u) = u/(1 - u)$. Uniform propensity learning and clipped overlap give conditional mean $O(a_{e,n} r_n)$ and conditional second moment $O(a_{e,n}^2 r_n)$. Consequently

$$\bar{g}_{d,n}(\theta_n) = \mathbb{P}_n \phi_{\theta_n} + o_P(\sqrt{r_n/n}), \quad \bar{\sigma}_{d,n}^2(\theta_n)/V_d^*(P_n) \rightarrow_P 1, \quad (2)$$

uniformly, because $nr_n a_{e,n}^2 \rightarrow 0$ and $nr_n \rightarrow \infty$. The bounded iid normal approximation applied to the oracle score then proves the studentized $N(0, 1)$ limit and hence uniform coverage; the fallback preserves coverage.

For diameter, the affine slope is $-\hat{r}_n = -\mathbb{P}_n U$, and $\sup_t \bar{\sigma}_{d,n}^2(t) \leq C_\varepsilon \mathbb{P}_n(U + J)$. Bounded exponential-moment bounds give, outside probability $Ce^{-c n r_n}$, $\hat{r}_n \geq r_n/2$ and $\mathbb{P}_n(U + J) \leq C_\varepsilon r_n$. Two accepted points are therefore separated by at most $C(nr_n)^{-1/2}$. The random set always has diameter at most one and the empty-set fallback has diameter zero, so expectation has the same order. This proves the two inference conclusions.

It remains to verify the advertised schedules. Since $s_e, s_b \in (0, 1/2)$ and $s_e + s_b < 1$, each of $1 - 2s_e$, $2s_b$, and $2s_e + 2s_b - 1$ is strictly below one; hence the displayed open interval for γ is nonempty. Eventually $r_n = n^{-\gamma}$, and, up to powers of $\log n$,

$$nr_n = n^{1-\gamma} \rightarrow \infty, \quad nr_n a_{e,n}^2 = n^{1-\gamma-2s_e} (\log n)^{2s_e} \rightarrow 0, \quad (3)$$

$$\frac{r_n}{a_{b,n}^2} = \frac{n^{2s_b-\gamma}}{(\log n)^{2s_b}} \rightarrow 0, \quad \frac{r_n}{na_{e,n}^2 a_{b,n}^2} = \frac{n^{2s_e+2s_b-1-\gamma}}{(\log n)^{2s_e+2s_b}} \rightarrow 0. \quad (4)$$

The cap by $\bar{d}$ changes only finitely many rows. Under LF compatibility and $c_G \leq 3/16$, `lem:boundary-shell-nonempty-witness` supplies nonemptiness. Equations (3)–(4) complete the robustness and boundary-refutation conclusions exactly as stated. $\square$

Justification. A propensity-only first-order inversion remains honest and shrinking on an explicit ultra-rare schedule class without outcome-regression smoothness. *Gap.* This is an internal estimator-robustness construction and does not assert that a published condition is necessary or false. *Consumer.* It provides a separate score option when the factual event is rare enough that $nr_n a_{e,n}^2 \rightarrow 0$ and the boundary shell can be verified.

Lemma 26 (`lem:tian-wu-regular-benchmark`). *With $m_a = \mu_a$, under strong ignorability, monotonicity, and positive factual denominators, Tian and Wu identify $\beta = \text{PN}$ and $\gamma = \text{PS}$ by*

$$\beta = 1 - \frac{E[e(X)m_0(X)]}{E[e(X)m_1(X)]}, \quad \gamma = \frac{E[(1 - e(X))\{m_1(X) - m_0(X)\}]}{E[(1 - e(X))(1 - m_0(X))]}.$$

Writing $q_{\text{PN}} = E[e(X)m_1(X)]$ and $q_{\text{PS}} = E[(1 - e(X))(1 - m_0(X))]$, their unknown-propensity monotonicity efficient influence functions are

$$\phi_{\text{PN}} = \frac{1 - \beta}{q_{\text{PN}}} AY - \frac{e(X)}{q_{\text{PN}}\{1 - e(X)\}} (1 - A)\{Y - m_0(X)\} - \frac{m_0(X)}{q_{\text{PN}}} A$$

and

$$\phi_{\text{PS}} = \frac{1 - e(X)}{q_{\text{PS}}e(X)} A\{Y - m_1(X)\} - \frac{1 - \gamma}{q_{\text{PS}}} (1 - A)\{Y - m_0(X)\} + \frac{m_0(X) - m_1(X) + \gamma\{1 - m_0(X)\}}{q_{\text{PS}}} (A - 1).$$

Their cross-fitted ratio estimators solve the corresponding empirical mean-zero influence-function equations; under their product-rate conditions, their regular inference theorems give root- n asymptotic normality, attainment of the respective efficiency bounds, and consistent empirical influence-function variance estimators.

Lemma 27 (`lem:cai-kuroki-regular-variance-context`). *Cai and Kuroki introduce the three probabilities of causation discussed by Pearl, derive variance estimators corresponding to Pearl's cases, clarify their properties, and report simulations showing that the proposed method sufficiently approximates the variances. This is regular variance-estimation context, not a rare-event or honest-diameter theorem.*

Theorem 8 (`thm:gaussian-frontier`). *Fix $\alpha \in (0, 1/2)$, $d \in \{\text{PN}, \text{PS}\}$, any fixed $c_G > 0$, set $C_G = \varepsilon^{-1}$, and let $(r_n) \in \mathcal{R}^d$. Suppose $\mathcal{M}_{G,n}^d(r_n; c_G, C_G) \neq \emptyset$ eventually. When $d = \text{PN}$, interpret every upper assertion below on $\mathcal{B}_n^{\text{PN}}$ using `prop:pn-causal-localization`, `lem:pn-variance-consistency`, and `lem:pn-structural-score-variance-upper-bound`; when $d = \text{PS}$, use the corresponding three PS nodes on $\mathcal{B}_n^{\text{PS}}$. No upper assertion imports smoothness of the regression belonging to the opposite direction. Uniformly over the matching shell,*

$$\hat{g}_{d,n}(\Theta_d(P_n)) = \frac{1}{n} \sum_{i=1}^n \psi_{\Theta_d(P_n),n}^d(O_{i,n}; \eta_n^d) + o_{P_n^{\otimes n}} \left(\sqrt{\frac{r_n}{n}} \right),$$

the studentized score converges uniformly to $N(0,1)$, its inversion has asymptotic coverage at least $1 - \alpha$, and

$$\sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} E_{P_n^{\otimes n}} [\text{diam}(\mathcal{C}_{n,\alpha}^{\text{score},d})] = O((nr_n)^{-1/2}).$$

These upper conclusions require no least-favorable compatibility. If the primitive tuple is least-favorable compatible, $0 < c_G \leq 3/16$, and $r_n \rightarrow 0$, every measurable $D_n \subseteq [0,1]$ satisfying

$$\liminf_{n \rightarrow \infty} \inf_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} P_n^{\otimes n} \{ \Theta_d(P_n) \in D_n \} \geq 1 - \alpha$$

obeys

$$\liminf_{n \rightarrow \infty} \sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} \sqrt{nr_n} E_{P_n^{\otimes n}} [\text{diam}(D_n)] \geq c_\alpha > 0.$$

Thus the score inversion has matched minimax expected-diameter order $(nr_n)^{-1/2}$ on each admissible vanishing-event identical shell under least-favorable compatibility. The theorem matches rates, not sharp constants, and asserts no converse for nonvanishing or oscillating schedules. The coordinate $C_G = \varepsilon^{-1}$ is derived rather than primitive. Cheng, Mao, Pearl, and Li's partially identified finite-discrete endpoint analysis concerns a different estimand and asymptotic regime.

Proof. Fix d , put $b = b(d)$, write $t_n = \Theta_d(P_n)$, and define

$$U_{i,n} = \psi_{t_n,n}^d(O_{i,n}; \eta_n^d), \quad V_n = V_d(P_n). \quad (1)$$

By `lem:identification`, $E_{P_n} U_{i,n} = 0$. The lower restriction in `def:gaussian-shells` and the matching structural score-variance upper bound give, uniformly over the direction-specific shell,

$$c_G r_n \leq V_n \leq \varepsilon^{-1} r_n. \quad (2)$$

For PN the upper inequality is `lem:pn-structural-score-variance-upper-bound`; for PS it is `lem:ps-structural-score-variance-upper-bound`. The schedule consequences are

$$nr_n \rightarrow \infty, \quad \frac{a_{b,n}^2}{r_n} \rightarrow 0, \quad \frac{na_{e,n}^2 a_{b,n}^2}{r_n} \rightarrow 0. \quad (3)$$

They follow from `lem:upper-schedule-consequences` and the definition of $\mathcal{R}^d$.

We first prove the oracle expansion. The finite-sample learner theorem `lem:finite-sample-directional-spline-learning-tails` supplies, outside probability $C_M n^{-M}$ and simultaneously over the fixed folds,

$$\|\hat{e} - e_n\|_\infty \vee \|\hat{e} - e_n\|_{2,P_n} \leq C_M a_{e,n}, \quad \|\hat{m}_b - m_{b,n}\|_\infty \vee \|\hat{m}_b - m_{b,n}\|_{2,P_n} \leq C_M a_{b,n}. \quad (4)$$

In the PN branch this statement involves only $m_{0,n}$; in the PS branch it involves only $m_{1,n}$. Let $\Delta_{i,n}$ be the fitted-minus-oracle held-out score difference at t_n . Direct subtraction in `def:scores`, conditional on the corresponding training fold, shows that its mean is a product of the two nuisance errors divided by a clipped overlap factor, and that

$$|E(\Delta_{i,n} \mid \text{training})| \leq C_\varepsilon a_{e,n} a_{b,n}, \quad E(\Delta_{i,n}^2 \mid \text{training}) \leq C_\varepsilon \{a_{b,n}^2 + a_{e,n}^2 h_{d,n}\}, \quad (5)$$

where $h_{\text{PN},n} = E[m_{0,n}(X)]$ and $h_{\text{PS},n} = E[1 - m_{1,n}(X)]$. The matching causal-localization proposition gives $h_{d,n} \leq r_n/\varepsilon$. Conditional Chebyshev on each held-out fold and a fixed-fold union bound therefore give

$$\hat{g}_{d,n}(t_n) - \frac{1}{n} \sum_{i=1}^n U_{i,n} = O_P \left(\frac{a_{b,n}}{\sqrt{n}} + a_{e,n} \sqrt{\frac{r_n}{n}} + a_{e,n} a_{b,n} \right) = o_P \left(\sqrt{\frac{r_n}{n}} \right) \quad (6)$$

uniformly over the shell. The last equality follows term by term from (3) and $a_{e,n} \rightarrow 0$. This proves the displayed oracle expansion.

For studentization, the matching variance-consistency lemma gives

$$\frac{\hat{\sigma}_{d,n}^2(t_n)}{V_n} \rightarrow_P 1 \quad (7)$$

uniformly over the shell. The oracle score is bounded by the common constant B_ε , and (2)–(3) give

$$\frac{B_\varepsilon}{\sqrt{nV_n}} \leq \frac{B_\varepsilon}{\sqrt{c_G n r_n}} \rightarrow 0. \quad (8)$$

Apply `lem:bounded-score-studentization` with $S_n = \hat{g}_{d,n}(t_n)$, oracle summands $U_{i,n}$, remainder from (6), and studentizer from (7). We obtain

$$\sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} \sup_{x \in \mathbb{R}} \left| P_n^{\otimes n} \left\{ \frac{\sqrt{n} \hat{g}_{d,n}(t_n)}{\hat{\sigma}_{d,n}(t_n)} \leq x \right\} - \Phi(x) \right| \rightarrow 0. \quad (9)$$

The event $\hat{\sigma}_{d,n}(t_n) = 0$ has probability tending to zero uniformly by (2) and (7). Evaluating (9) at $\pm z_{1-\alpha/2}$ proves

$$\inf_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} P_n^{\otimes n} \{ n \hat{g}_{d,n}(t_n)^2 \leq z_{1-\alpha/2}^2 \hat{\sigma}_{d,n}^2(t_n) \} \rightarrow 1 - \alpha. \quad (10)$$

The event in (10) is $\{t_n \in \mathcal{A}_{n,\alpha}^d\}$. On this event the preliminary set is nonempty, so the fallback rule in `def:score-inversion` leaves it unchanged. Hence

$$\liminf_{n \rightarrow \infty} \inf_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} P_n^{\otimes n} \{t_n \in \mathcal{C}_{n,\alpha}^{\text{score},d}\} \geq 1 - \alpha. \quad (11)$$

We next prove the expected-diameter upper bound. Define

$$H_d(O) = \begin{cases} AY, & d = \text{PN}, \\ (1-A)(1-Y), & d = \text{PS}, \end{cases} \quad \hat{q}_{d,n} = \mathbb{P}_n H_d. \quad (12)$$

On the shell, $EH_d = r_n$. Since $H_d \in [0, 1]$ and $\text{Var}(H_d) \leq r_n$, `lem:bounded-bernstein-row-concentration`, with $x = cnr_n$ for a sufficiently small fixed $c > 0$, yields

$$\sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} P_n^{\otimes n} \{ \hat{q}_{d,n} < r_n/2 \} \leq 2e^{-cnr_n}. \quad (13)$$

By `lem:rare-score-empirical-envelope`, for any fixed $M > 0$, there are constants $C_M, c_M > 0$ such that

$$P_n^{\otimes n} \left\{ \sup_{t \in [0,1]} \mathbb{P}_n [\psi_{t,n}^d(O; \hat{\eta}_{d,-k,n})^2] > C_M r_n \right\} \leq C_M n^{-M} + C_M e^{-c_M n r_n} \quad (14)$$

uniformly over the shell. Let $\mathcal{E}_n$ be the intersection of the complementary event in (14) with $\{\hat{q}_{d,n} \geq r_n/2\}$. Taking $M = 2$, equations (13)–(14) give

$$\sup_{P_n} P_n^{\otimes n} (\mathcal{E}_n^c) \leq Cn^{-2} + Ce^{-cnr_n}. \quad (15)$$

If $t \in \mathcal{A}_{n,\alpha}^d$, then $\hat{\sigma}_{d,n}^2(t) \leq \mathbb{P}_n [\psi_{t,n}^d(O; \hat{\eta}_{d,-k,n})^2]$. Therefore, on $\mathcal{E}_n$,

$$|\hat{g}_{d,n}(t)| \leq z_{1-\alpha/2} \sqrt{\frac{Cr_n}{n}}. \quad (16)$$

The score is affine in t , with exact slope $-\hat{q}_{d,n}$:

$$\hat{g}_{d,n}(t) - \hat{g}_{d,n}(s) = -(t-s)\hat{q}_{d,n}. \quad (17)$$

Thus any $s, t \in \mathcal{A}_{n,\alpha}^d$ satisfy, on $\mathcal{E}_n$,

$$|t-s| \leq \frac{4z_{1-\alpha/2}\sqrt{C}}{\sqrt{nr_n}}. \quad (18)$$

The fallback is a singleton when the preliminary set is empty, and every reported set is a subset of $[0, 1]$. Hence

$$\text{diam}(\mathcal{C}_{n,\alpha}^{\text{score},d}) \leq \frac{4z_{1-\alpha/2}\sqrt{C}}{\sqrt{nr_n}} \mathbf{1}_{\mathcal{E}_n} + \mathbf{1}_{\mathcal{E}_n^c}. \quad (19)$$

Because $r_n \leq 1$, one has $n^{-2} = O((nr_n)^{-1/2})$. Since $nr_n \rightarrow \infty$, also $e^{-c nr_n} = O((nr_n)^{-1/2})$. Taking expectations in (19) and using (15) proves

$$\sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} E_{P_n^{\otimes n}}[\text{diam}(\mathcal{C}_{n,\alpha}^{\text{score},d})] = O((nr_n)^{-1/2}). \quad (20)$$

All upper-bound arguments were branch-specific: PN used only the PN learner, m_0 -localization, PN variance consistency, and the PN structural variance bound, while PS used only the corresponding objects involving m_1 . No least-favorable compatibility and no opposite-direction regression smoothness were used.

Finally, suppose the primitive tuple is least-favorable compatible, $0 < c_G \leq 3/16$, and $r_n \rightarrow 0$. Then $r_n \leq \bar{d}$ eventually and $(c_G, \varepsilon^{-1}) \in \mathcal{K}_{\text{LF}}$. The lemma `lem:lf-membership` places its local two-point pair in the identical direction-specific shell. Applying `conj:gaussian-lower` to any measurable set D_n satisfying the displayed honesty requirement gives

$$\liminf_{n \rightarrow \infty} \sup_{P_n \in \mathcal{M}_{G,n}^d(r_n; c_G, C_G)} \sqrt{nr_n} E_{P_n^{\otimes n}}[\text{diam}(D_n)] \geq c_\alpha > 0. \quad (21)$$

The single data-driven score inversion attaining (20) therefore matches (21) in order. The argument identifies no sharp constant and uses eventual capping through $r_n \rightarrow 0$, so it gives no converse for nonvanishing or oscillating schedules. The equality $C_G = \varepsilon^{-1}$ records the proved structural upper coefficient and is not an additional primitive restriction. Finally, `lem:cheng-mao-pearl-li-2026-comparator` concerns partially identified finite-discrete endpoints, so it neither changes the estimand nor supplies any step in the identical-shell frontier proved here. $\square$

Justification. The mathematical closure is direction-specific: PN uses only PN localization, variance consistency, and structural variance, while PS uses only the corresponding PS nodes. The closest-work distinction has four axes. First, on identification, Zhang, Wang, and Miao [4] study inference for generally partially identified PN bound endpoints, whereas Cuellar and Kennedy [10] impose identification conditions including monotonicity but target a finite-dimensional projection of covariate-specific PC; the present theorem takes the exact marginal PN or PS functional under ignorability and monotonicity as given. Second, on scale, both cited analyses establish ordinary root-sample-size asymptotics, while the present theorem is a triangular-array result with $q_d(P_n) = r_n \rightarrow 0$, score variance of order r_n , and effective count nr_n . Third, the estimator is an affine cross-fitted PN or PS score-test inversion. Fourth, neither cited source states an honest expected-diameter lower bound over an identical vanishing-event shell. The achievability proof derives the spline rates from explicit finite-sample concentration and a proved inverse-row-sum inequality rather than treating learner rates as black-box assertions. *Gap.* The matched result is rate-only and its converse is restricted to admissible vanishing schedules under least-favorable compatibility. It does not identify a sharp constant or supply a converse for nonvanishing or oscillating schedules. The bounded-count result remains a lower obstruction rather than a Poisson-calibrated upper procedure. *Consumer.* Climate PN/FAR and stroke PS users may apply only the matching direction after verifying the factual schedule, score-information lower bound, overlap, directional Hölder conditions, and the stated effective-count regime.

Lemma 28 (`lem:tensor-spline-inverse-row-sum`). *Let A_J be a real symmetric positive-definite matrix indexed by a finite subset $\Lambda_J \subset \mathbb{Z}^p$. Suppose fixed constants $0 < a \leq b < \infty$, $R < \infty$, and $D < \infty$ satisfy $\text{spec}(A_J) \subset [a, b]$, $(A_J)_{\lambda\mu} = 0$ when $\|\lambda - \mu\|_\infty > R$, and, for the graph induced by the nonzero off-diagonal entries, every distance shell obeys*

$$\sup_{\lambda \in \Lambda_J} \#\{\mu \in \Lambda_J : \text{dist}(\lambda, \mu) = s\} \leq D(s+1)^{p-1}, \quad s \geq 0.$$

Then $\|A_J^{-1}\|_{\infty \rightarrow \infty} \leq C(a, b, D, p)$, uniformly in J . Consequently, if H_J has the same sparsity and shell-growth properties and $\text{spec}(H_J) \subset [a/J, b/J]$, then

$$\|H_J^{-1}\|_{\infty \rightarrow \infty} \leq C(a, b, D, p)J.$$

Proof. Put $c = 2/(a+b)$, $\rho = (b-a)/(a+b) < 1$, and $R_J = I - cA_J$. Spectral calculus gives

$$\|R_J\|_{2 \rightarrow 2} \leq \rho, \quad A_J^{-1} = c \sum_{k=0}^{\infty} R_J^k. \quad (1)$$

The diagonal entries of R_J add only self-loops to the graph induced by the nonzero off-diagonal entries of A_J . Consequently, if $h = \text{dist}(\lambda, \mu)$, then $(R_J^k)_{\lambda\mu} = 0$ for every $k < h$. Moreover,

$$|(R_J^k)_{\lambda\mu}| \leq \|R_J^k\|_{2 \rightarrow 2} \leq \rho^k. \quad (2)$$

Combining (1) and (2) yields

$$|(A_J^{-1})_{\lambda\mu}| \leq c \sum_{k=h}^{\infty} \rho^k = \frac{c}{1-\rho} \rho^h. \quad (3)$$

For a fixed row λ , sum (3) over graph-distance shells and use the assumed shell-growth bound. Then

$$\sum_{\mu \in \Lambda_J} |(A_J^{-1})_{\lambda\mu}| \leq \frac{cD}{1-\rho} \sum_{s=0}^{\infty} (s+1)^{p-1} \rho^s =: C(a, b, D, p) < \infty. \quad (4)$$

The series in (4) converges because $0 \leq \rho < 1$. Taking the supremum over λ proves $\|A_J^{-1}\|_{\infty \rightarrow \infty} \leq C(a, b, D, p)$, uniformly in J . For the final assertion, apply the first part to $A_J = JH_J$. Its spectrum lies in $[a, b]$, and its sparsity graph is unchanged. Since $H_J^{-1} = JA_J^{-1}$,

$$\|H_J^{-1}\|_{\infty \rightarrow \infty} \leq C(a, b, D, p)J,$$

as required. $\square$

Lemma 29 (lem:bounded-bernstein-row-concentration). *Let $Z_1, \dots, Z_N$ be independent real random variables with $EZ_i = 0$, $|Z_i| \leq 1$ almost surely, and $N^{-1} \sum_{i=1}^N EZ_i^2 \leq \nu$. Then, for every $x > 0$,*

$$P \left\{ \left| N^{-1} \sum_{i=1}^N Z_i \right| > \sqrt{\frac{2\nu x}{N}} + \frac{2x}{3N} \right\} \leq 2e^{-x}.$$

Proof. For $0 < \lambda < 3$, the conditions $|Z_i| \leq 1$ and $EZ_i = 0$, together with $k! \geq 2 \cdot 3^{k-2}$ for $k \geq 2$, imply

$$Ee^{\lambda Z_i} = 1 + \sum_{k=2}^{\infty} \frac{\lambda^k EZ_i^k}{k!} \leq 1 + \frac{\lambda^2 EZ_i^2}{2(1-\lambda/3)}. \quad (1)$$

Using $\log(1+u) \leq u$, independence, and $\sum_i EZ_i^2 \leq N\nu$, we obtain

$$\log E \exp \left(\lambda \sum_{i=1}^N Z_i \right) \leq \frac{N\nu\lambda^2}{2(1-\lambda/3)}. \quad (2)$$

Markov's inequality applied to the exponential in (2), followed by minimization over $0 < \lambda < 3$, gives the one-sided Bernstein inequality

$$P \left\{ \sum_{i=1}^N Z_i > \sqrt{2N\nu x} + \frac{2x}{3} \right\} \leq e^{-x}. \quad (3)$$

For completeness, if $\nu > 0$, substitution of

$$\lambda = \frac{\sqrt{2x/(N\nu)}}{1 + \sqrt{2x/(9N\nu)}}$$

in the Chernoff bound from (2) gives (3) after elementary simplification; if $\nu = 0$, every Z_i vanishes almost surely and (3) is immediate. Apply (3) also to $-Z_i$, add the two tail probabilities, and divide the threshold by N . This gives

$$P \left\{ \left| N^{-1} \sum_{i=1}^N Z_i \right| > \sqrt{\frac{2\nu x}{N}} + \frac{2x}{3N} \right\} \leq 2e^{-x},$$

which is the claimed result. $\square$

Lemma 30 (lem:finite-sample-directional-spline-learning-tails). *Fix $d \in \{\text{PN}, \text{PS}\}$, put $b = b(d)$, and use the direction- d learner from **def:pn-learners** or **def:ps-learners**. Under row-wise sampling and the assumptions defining the matching directional upper class, for every fixed $M > 0$ there is $C_M < \infty$, depending only on the frozen primitive constants and M , such that, uniformly over that upper class and all sufficiently large n ,*

$$P_n^{\otimes n} \left\{ \max_{1 \leq k \leq K} \left(\frac{\|\widehat{e}_{-k,n} - e_n\|_\infty}{a_{e,n}} \vee \frac{\|\widehat{m}_{b,-k,n} - m_{b,n}\|_\infty}{a_{b,n}} \right) > C_M \right\} \leq C_M n^{-M}.$$

For $d = \text{PN}$ the second term uses only $m_{0,n}$; for $d = \text{PS}$ it uses only $m_{1,n}$. In particular, the same bounds hold in $L_2(P_n)$, and no smoothness condition on the regression belonging to the opposite direction is used.

Proof. Fix a direction d , put $b = b(d)$, and fix one training fold. Write N for its size. Balanced fixed-fold splitting gives constants $0 < c_K < C_K < \infty$ such that $c_K n \leq N \leq C_K n$ for all sufficiently large n . We first establish the deterministic spline bounds and the common high-probability event.

For a degree- r tensor-product B-spline vector $B(x)$ on a quasi-uniform grid with κ cells per coordinate and $J = (\kappa + r + 1)^p$, nonnegativity and partition of unity give

$$0 \leq B_j(x) \leq 1, \quad \sum_{j=1}^J B_j(x) = 1. \quad (1)$$

Each basis function overlaps only a fixed number of other basis functions. The overlap graph is a fixed-range graph on a subset of $\mathbb{Z}^p$, and its graph-distance shell of radius s has at most $D_{r,p}(s+1)^{p-1}$ vertices. On each grid cell, affine rescaling to the unit cube turns the active splines into a fixed-dimensional polynomial basis with a positive-definite Gram matrix. Quasi-uniformity restricts all rescaled knot configurations to a compact set. The smallest and largest eigenvalues of the local Gram matrices are therefore bounded away from zero and infinity uniformly in κ . Summing the local inequalities over cells and using the support-wide bounds on $f_{X,n}$ yields constants $0 < c < C < \infty$, depending only on the frozen primitives, such that

$$cJ^{-1}\|v\|_2^2 \leq E\{B(X)^\top v\}^2 \leq CJ^{-1}\|v\|_2^2, \quad EB_j(X) \leq C/J. \quad (2)$$

By overlap, multiplication by $P(A = a \mid X)$, for $a \in \{0, 1\}$, preserves (2) up to constants.

The local Taylor formula through order $\lceil \beta \rceil - 1$, the Hölder remainder in **def:holder**, and the fixed-degree local spline dual functionals construct a tensor quasi-interpolant $B^\top \theta^\circ$ for every $f \in \mathcal{H}^\beta(L)$ satisfying

$$\|B^\top \theta^\circ - f\|_\infty \leq C\kappa^{-\beta}, \quad \|\theta^\circ\|_\infty \leq C. \quad (3)$$

Indeed, the Taylor remainder on a cell is at most $C\kappa^{-\beta}$; the local dual functionals reproduce its Taylor polynomial and have uniformly bounded operator norms by compactness of the rescaled knot configurations; and the fixed-overlap partition of unity combines the cellwise bounds. For the propensity, let $\ell_n = \log\{e_n/(1 - e_n)\}$. Overlap and the chain rule bound the full Hölder norm of ℓ_n uniformly. The calibrated quasi-interpolation rule encoded by B_e in **def:learners** therefore provides θ_e° such that

$$\|B^\top \theta_e^\circ - \ell_n\|_\infty \leq C\kappa^{-\beta_e}, \quad \|\theta_e^\circ\|_\infty \leq B_e - 1. \quad (4)$$

We now construct a finite-sample event. Every nonzero coordinate of $B_j B_l$, $\mathbf{1}\{A = a\} B_j B_l$, $B_j\{A - e_n(X)\}$, and $\mathbf{1}\{A = a\} B_j\{Y - m_{a,n}(X)\}$ is bounded by one and has second moment at most C/J , by (1)–(2). There are at most $D_r J$ nonzero Gram coordinates and at most J score coordinates. Apply **lem:bounded-bernstein-row-concentration** with $x = (M + 4) \log n$, and take a union bound over these coordinates and the fixed number of folds. With

$$u_{M,n} = C_M \left\{ \sqrt{\frac{\log n}{nJ}} + \frac{\log n}{n} \right\}, \quad (5)$$

there is an event $\mathcal{E}_{M,n}$ such that

$$\sup_{P_n} P_n^{\otimes n}(\mathcal{E}_{M,n}^c) \leq C_M n^{-M}, \quad (6)$$

and, on this event, every preceding empirical coordinate differs from its expectation by at most $u_{M,n}$. A further coordinatewise application gives $\max_j \mathbb{P}_N B_j \leq C_M/J$, since

$$Ju_{M,n} \leq C_M \left\{ \sqrt{\frac{J \log n}{n}} + \frac{J \log n}{n} \right\} = o(1). \quad (7)$$

Let $G_N = \mathbb{P}_N[BB^\top]$ and $G_{a,N} = \mathbb{P}_N[\mathbf{1}\{A = a\}BB^\top]$. Each row of an empirical-minus-population Gram matrix has only a fixed number of nonzero entries. Thus (2), (5), (7), and the maximum-row-sum bound for the spectral norm imply, on $\mathcal{E}_{M,n}$,

$$cJ^{-1}I_J \preceq G_N \preceq CJ^{-1}I_J, \quad cJ^{-1}I_J \preceq G_{a,N} \preceq CJ^{-1}I_J. \quad (8)$$

Because $\lambda = n^{-2}$, the scaled matrices $J(G_{a,N} + \lambda I_J)$ have spectra in a fixed compact subset of $(0, \infty)$ for all sufficiently large n . They have the fixed-range tensor-lattice sparsity and polynomial shell growth required by `lem:tensor-spline-inverse-row-sum`. Consequently,

$$\|(G_{a,N} + \lambda I_J)^{-1}\|_{\infty \rightarrow \infty} \leq CJ. \quad (9)$$

Consider the outcome learner for the matching arm $a = b$, and choose θ_a° as in (3). Its ridge normal equation is

$$(G_{a,N} + \lambda I_J)(\hat{\theta}_a - \theta_a^\circ) = \mathbb{P}_N[\mathbf{1}\{A = a\}B\{Y - m_{a,n}(X)\}] + \mathbb{P}_N[\mathbf{1}\{A = a\}B\{m_{a,n}(X) - B^\top \theta_a^\circ\}] - \lambda \theta_a^\circ. \quad (10)$$

On $\mathcal{E}_{M,n}$, the infinity norm of the right side is at most

$$u_{M,n} + C\kappa^{-\beta_a}/J + C\lambda. \quad (11)$$

Equations (3), (9)–(11), and partition of unity give

$$\|B^\top \hat{\theta}_a - m_{a,n}\|_\infty \leq C_M \left\{ \kappa^{-\beta_a} + \sqrt{\frac{J \log n}{n}} + \frac{J \log n}{n} + \frac{J}{n^2} \right\}. \quad (12)$$

For the declared $\kappa = K_{a,n}$, one has $J \asymp \kappa^p$, and direct substitution shows

$$\kappa^{-\beta_a} + \sqrt{\frac{J \log n}{n}} + \frac{J \log n}{n} + \frac{J}{n^2} \leq C \left\{ \frac{\log n}{n} \right\}^{\beta_a/(2\beta_a+p)} = Ca_{a,n}. \quad (13)$$

Pointwise projection onto $[0, 1]$ is contractive toward the $[0, 1]$ -valued true regression, so (12)–(13) prove the asserted outcome-regression bound. Only $m_{0,n}$ is used in the PN branch, and only $m_{1,n}$ is used in the PS branch.

It remains to treat the propensity. Put $p_\theta(x) = \{1 + \exp(-B(x)^\top \theta)\}^{-1}$, let L_N denote the ridge-logistic objective, and set $s_N = \nabla L_N(\theta_e^\circ)$. The sigmoid is $1/4$ -Lipschitz. Hence (4), (5), and the local-mass bound imply

$$\|s_N\|_\infty \leq u_{M,n} + C\kappa^{-\beta_e}/J + Cn^{-2}, \quad J\|s_N\|_\infty \leq C_M a_{e,n}. \quad (14)$$

Choose a fixed R_M larger than the constant in the second bound. For $\|\delta\|_\infty \leq R_M a_{e,n}$, define the integrated Hessian

$$\overline{H}_N(\delta) = \int_0^1 \nabla^2 L_N(\theta_e^\circ + u\delta) du. \quad (15)$$

For large n , $R_M a_{e,n} < 1/2$. By (4), the entire path in (15) lies strictly inside the coefficient cube, and $|B^\top(\theta_e^\circ + u\delta)| \leq B_e$. The logistic weights on the path are therefore between

$$w_\star = \frac{\exp(B_e)}{\{1 + \exp(B_e)\}^2} > 0 \quad \text{and} \quad \frac{1}{4}. \quad (16)$$

Equations (8) and (16) show that $J\overline{H}_N(\delta)$ has spectrum in a fixed compact subinterval of $(0, \infty)$, uniformly over the displayed cube. It has the same fixed-range sparsity as G_N . The inverse-row-sum lemma gives

$$\sup_{\|\delta\|_\infty \leq R_M a_{e,n}} \|\overline{H}_N(\delta)^{-1}\|_{\infty \rightarrow \infty} \leq CJ. \quad (17)$$

Define $T_N(\delta) = -\overline{H}_N(\delta)^{-1}s_N$. Equations (14) and (17) show that T_N maps the closed cube $\{\|\delta\|_\infty \leq R_M a_{e,n}\}$ into itself. The derivative of the logistic weight is bounded, while partition of unity and the empirical local-mass bound give

$$\|\overline{H}_N(\delta_1) - \overline{H}_N(\delta_2)\|_{\infty \rightarrow \infty} \leq CJ^{-1}\|\delta_1 - \delta_2\|_\infty. \quad (18)$$

Using the inverse identity in (18), together with (14) and (17), yields

$$\|T_N(\delta_1) - T_N(\delta_2)\|_\infty \leq Ca_{e,n}\|\delta_1 - \delta_2\|_\infty \leq \frac{1}{2}\|\delta_1 - \delta_2\|_\infty \quad (19)$$

for all sufficiently large n . Iteration from zero therefore converges to a fixed point δ_N^* in the cube. The fundamental theorem of calculus and the fixed-point equation give

$$abla L_N(\theta_e^\circ + \delta_N^*) = s_N + \bar{H}_N(\delta_N^*)\delta_N^* = 0. \quad (20)$$

The positive ridge makes the unrestricted objective strictly convex and coercive. Thus $\theta_e^\circ + \delta_N^*$ is its unique minimizer; since it is strictly inside the coefficient cube, it is also the constrained minimizer in `def:learners`. Partition of unity, (4), and sigmoid Lipschitz continuity now yield

$$\|p_{\hat{\theta}_e} - e_n\|_\infty \leq \frac{1}{4}\|\delta_N^*\|_\infty + C\kappa^{-\beta_e} \leq C_M a_{e,n}. \quad (21)$$

Projection onto $[\varepsilon/2, 1 - \varepsilon/2]$ is contractive toward e_n , so (21) also holds for $\hat{e}_{-k,n}$. Combining (6), (12)–(13), and (21), and retaining the fixed-fold union bound, proves the displayed probability inequality uniformly over the matching directional upper class. Supremum-norm control implies the asserted $L_2(P_n)$ control. Every constant used only the frozen overlap, density, Hölder, degree, dimension, ridge, and fold constants. In particular, no smoothness property of the opposite-direction regression entered either branch. $\square$

7 Interpretation

The factual conditioning event supplies effective count nr_n . Once the directional spline errors are proved smaller than the score scale, the affine score mean fluctuates at order $\sqrt{r_n/n}$; division by factual slope r_n gives diameter $(nr_n)^{-1/2}$. The direct inverse-row-sum argument is what converts coordinatewise empirical concentration into the required uniform fitted-function rates. The identical-shell least-favorable pair proves that the resulting diameter order cannot be uniformly improved. When nr_n is bounded, separated attribution values remain statistically close and honest sets cannot shrink.

Technical internal limitation (diagnostic only)

The Gaussian upper theorem requires $r_n/\{na_{e,n}^2 a_{b(d),n}^2\} \rightarrow \infty$, eventual shell nonemptiness, and $V_d(P_n) \geq c_G r_n$. The matching converse additionally requires least-favorable compatibility, $c_G \leq 3/16$, and a vanishing eventually capped schedule, and it matches rates rather than constants. The finite-sample learner argument uses fixed dimension p , fixed spline degree, fixed fold count, quasi-uniform tensor grids, bounded outcomes, and support-wide design-density bounds. The bounded-count theorem remains a lower obstruction rather than a Poisson-calibrated upper procedure.

8 Honest scope

The contribution is inference for already identified PN/PS, not new identification, a new efficient influence function, a new cross-fitting principle, or a new generic inverse-decay method. The headline is the conjunction of a known drifting schedule, a nonlinear causal ratio, an identical-shell matched expected-diameter order, and the bounded-effective-count obstruction. The matrix and concentration arguments complete the explicit estimator’s achievability proof but create no separate novelty claim. The climate and stroke examples remain consumers rather than demonstrations of shell membership.

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